

BFS Traversal

Step 1: Start at E

Legend

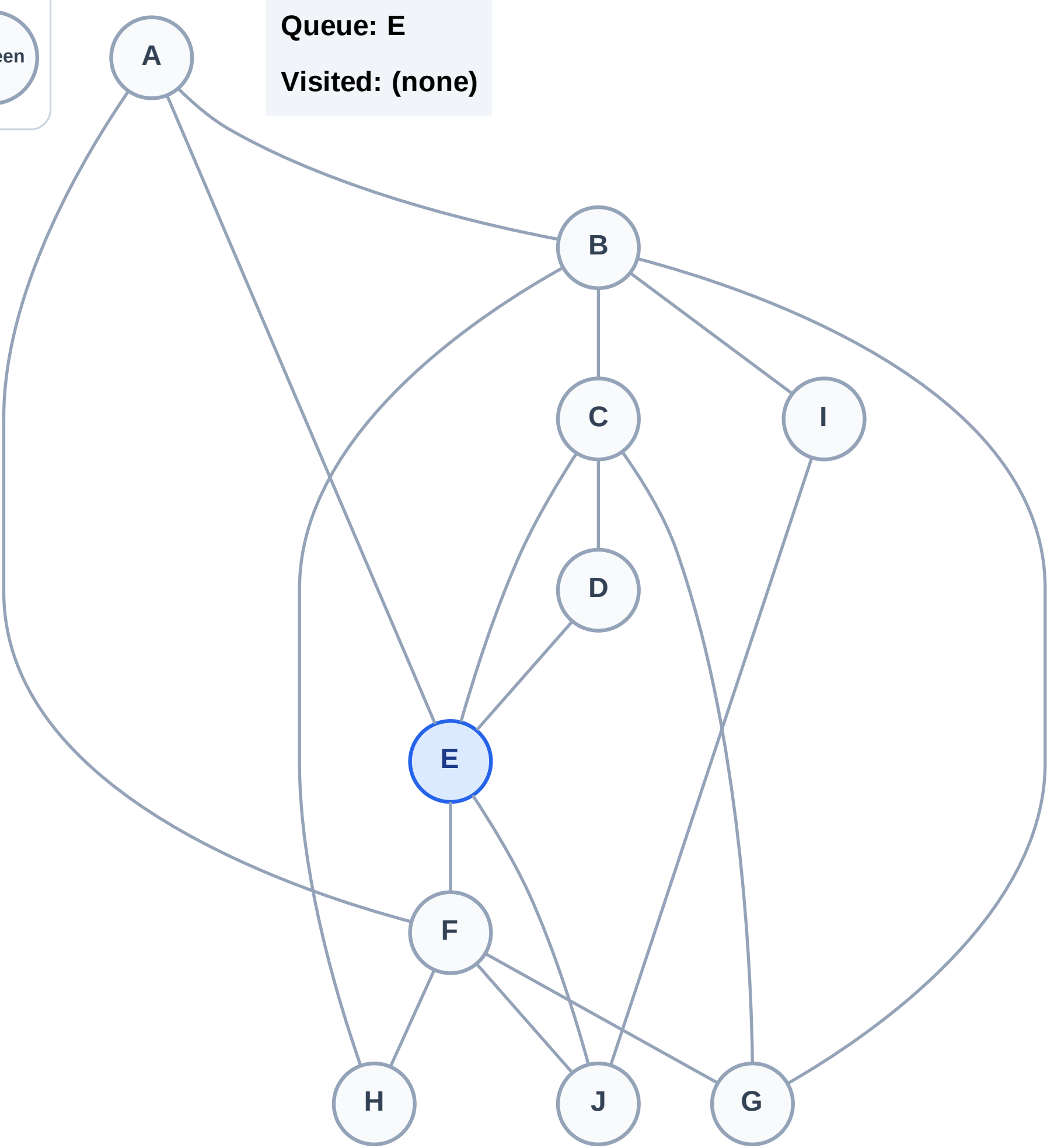
Complete

Processing

Discovered

Unseen

Queue: E
Visited: (none)



BFS Traversal

Step 2: Process node E

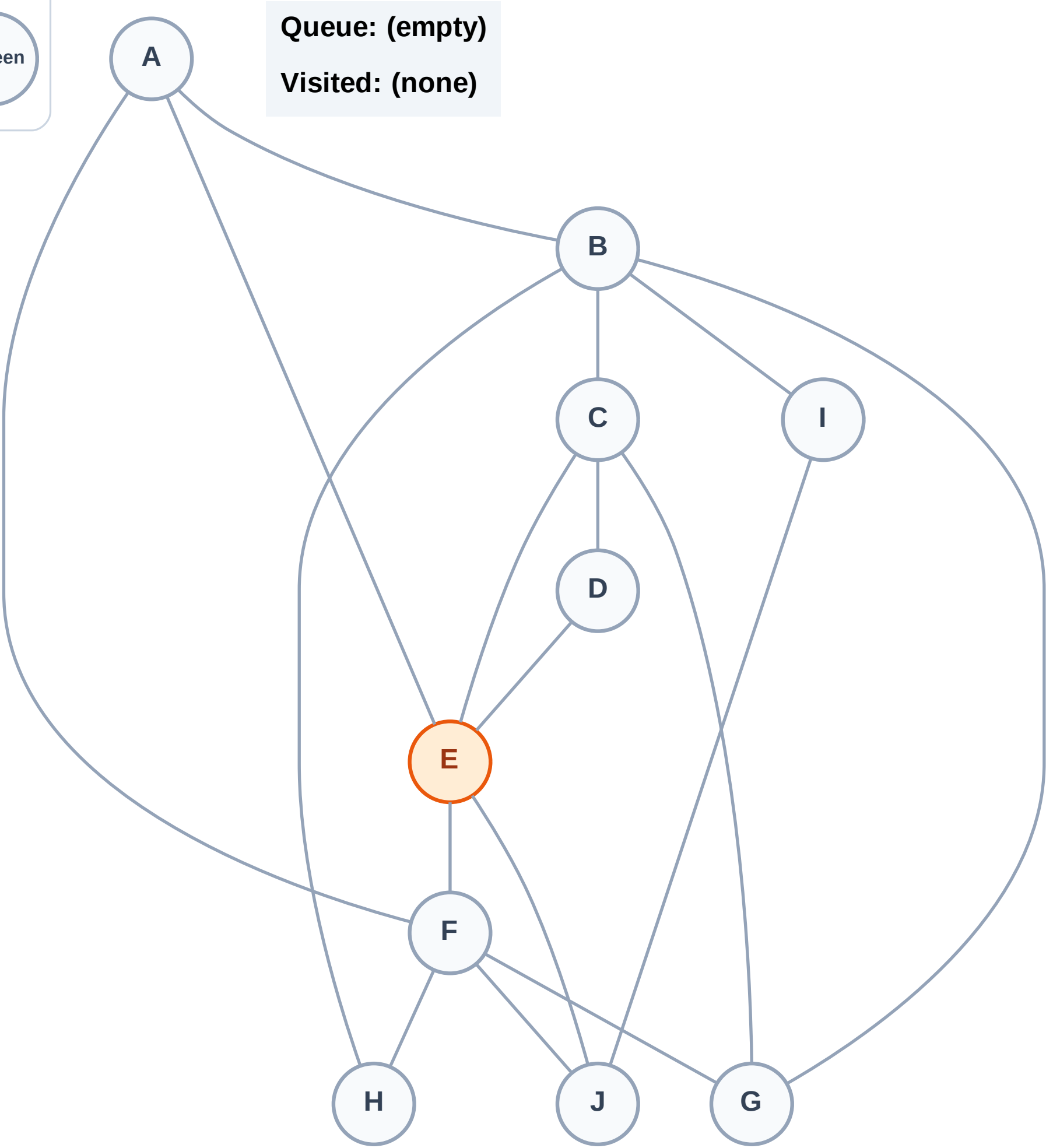
Legend

Complete

Processing

Discovered

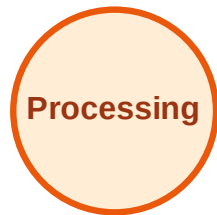
Unseen



BFS Traversal

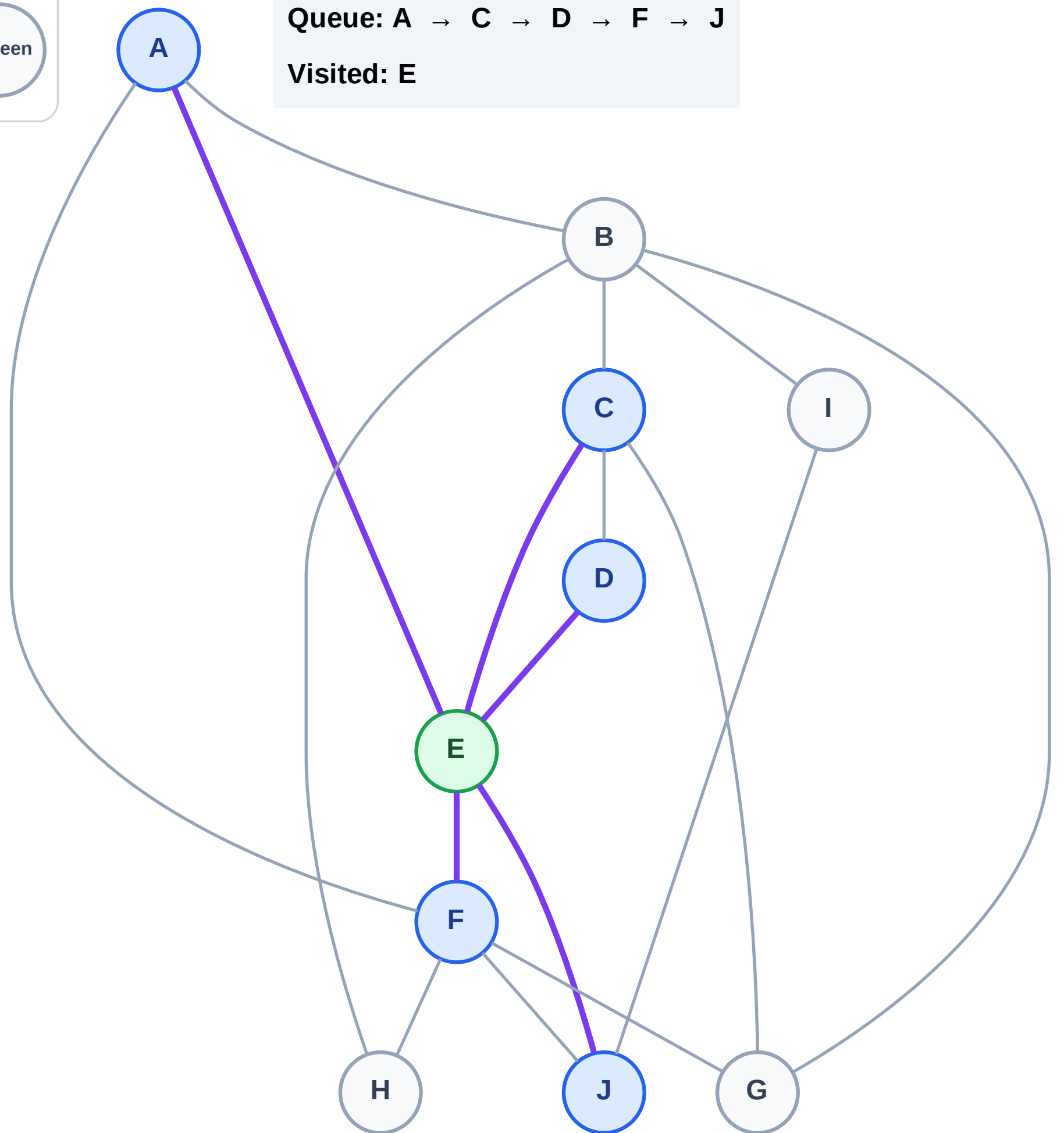
Step 3: Discover A, C, D, F, J from E

Legend



Queue: A → C → D → F → J

Visited: E



BFS Traversal

Step 4: Process node A

Legend

Complete

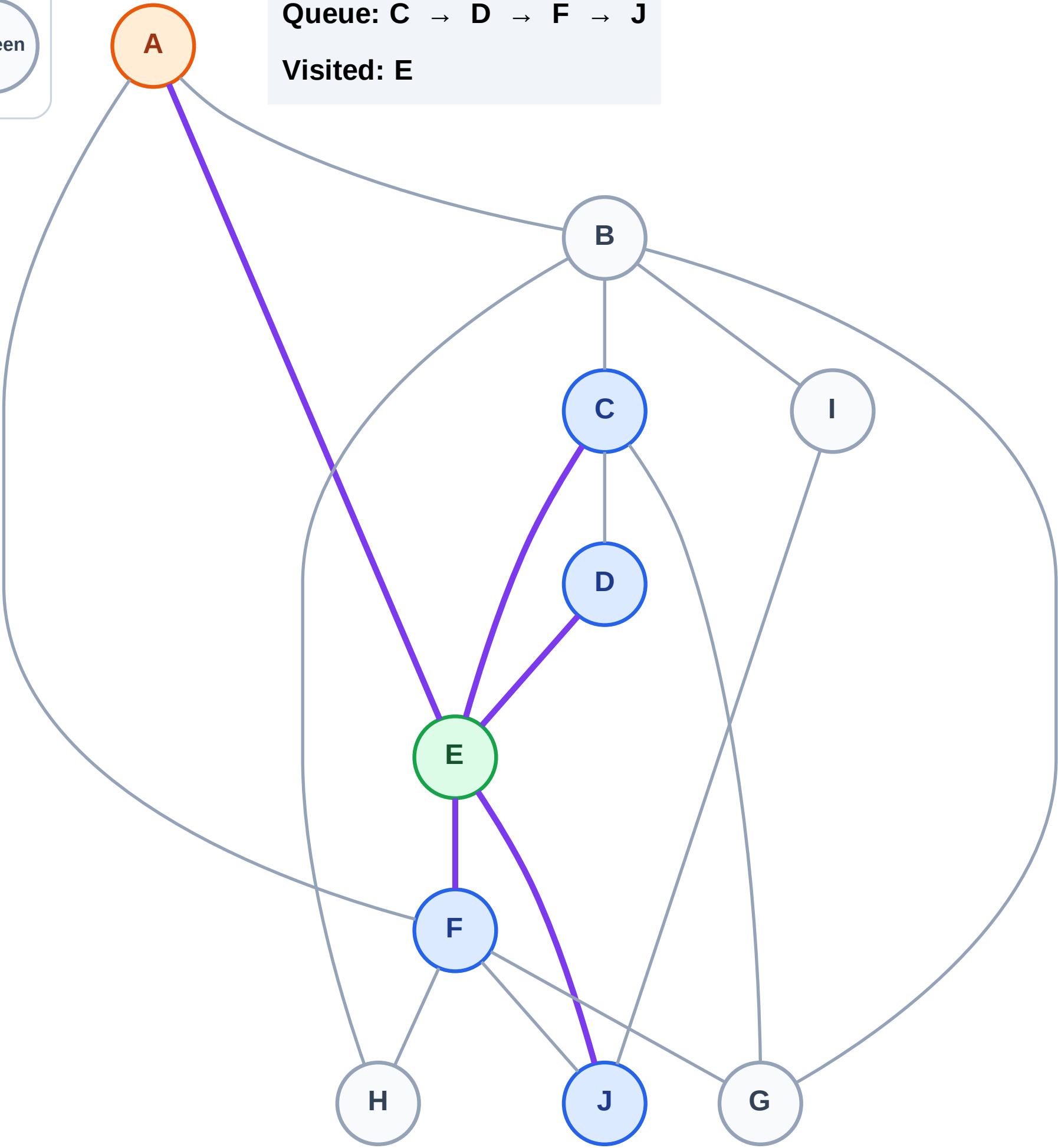
Processing

Discovered

Unseen

Queue: C → D → F → J

Visited: E



BFS Traversal

Step 5: Discover B from A

Legend



Complete



Processing



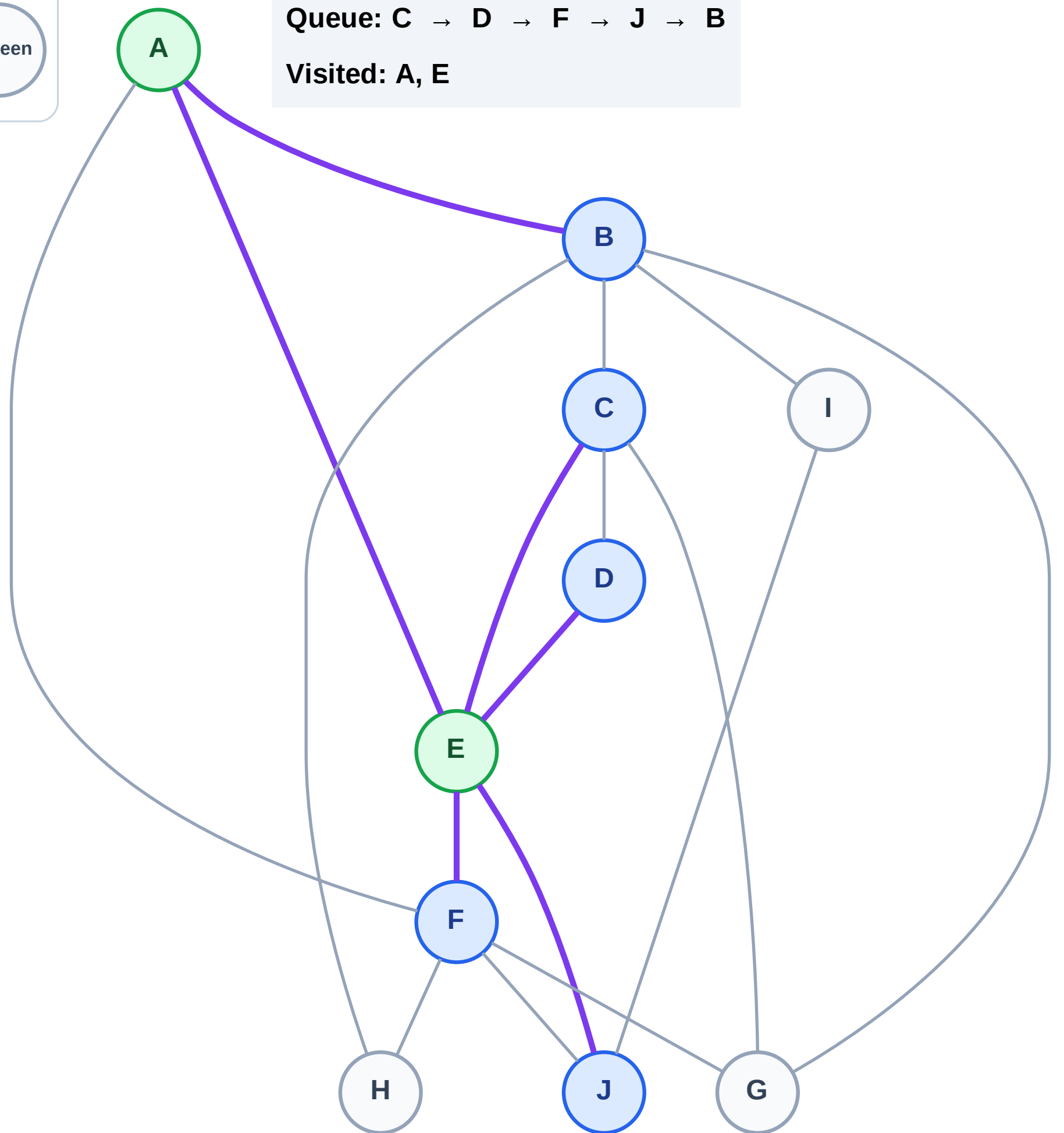
Discovered



Unseen

Queue: C → D → F → J → B

Visited: A, E



BFS Traversal

Step 6: Process node C

Legend

Complete

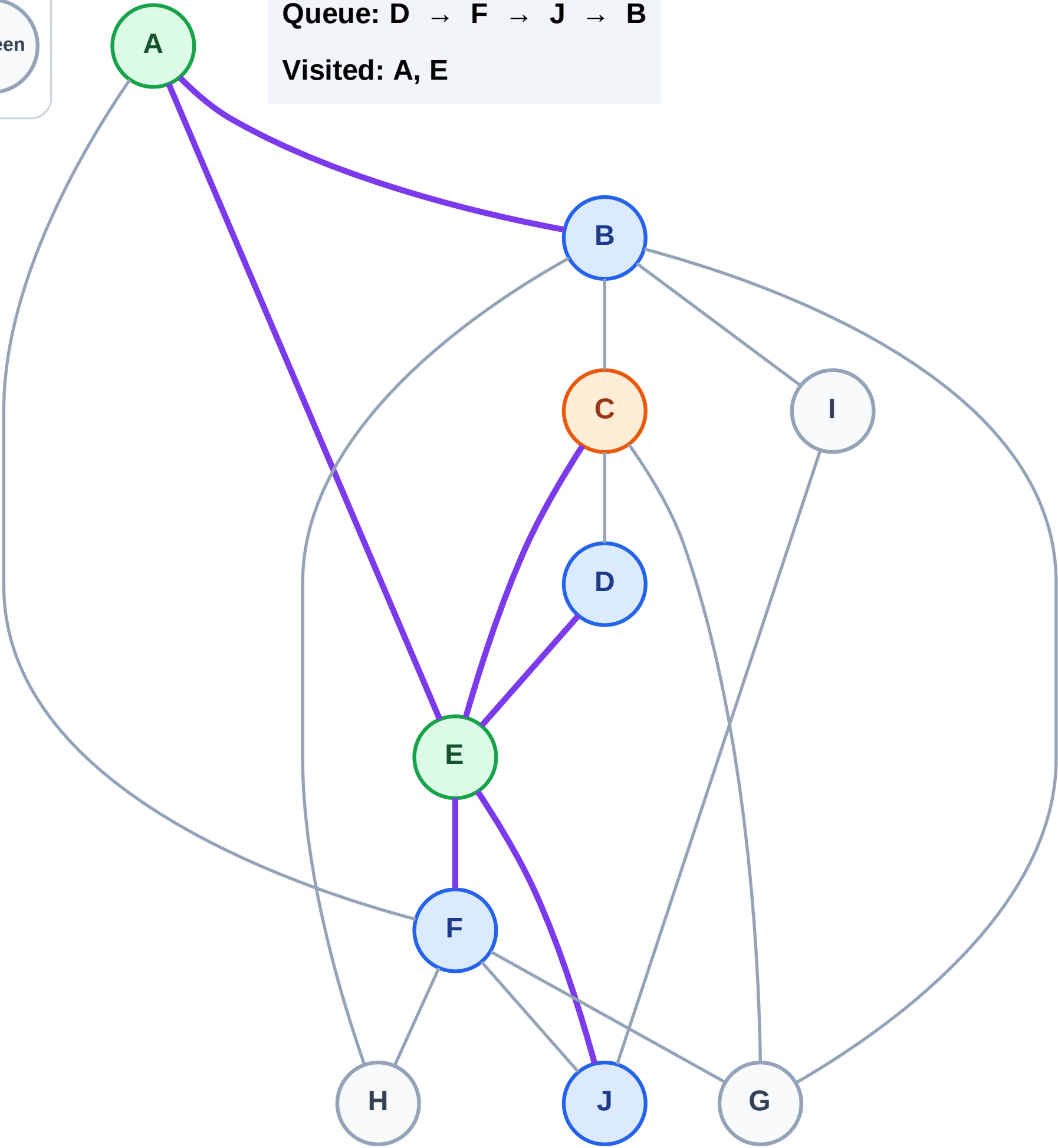
Processing

Discovered

Unseen

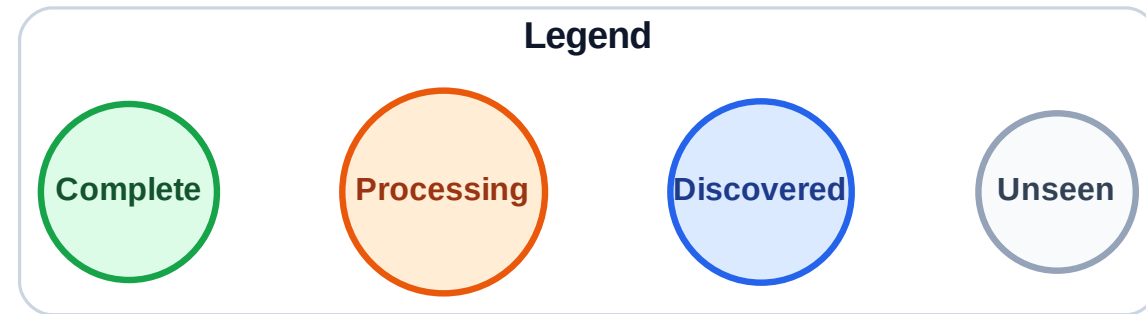
Queue: D → F → J → B

Visited: A, E



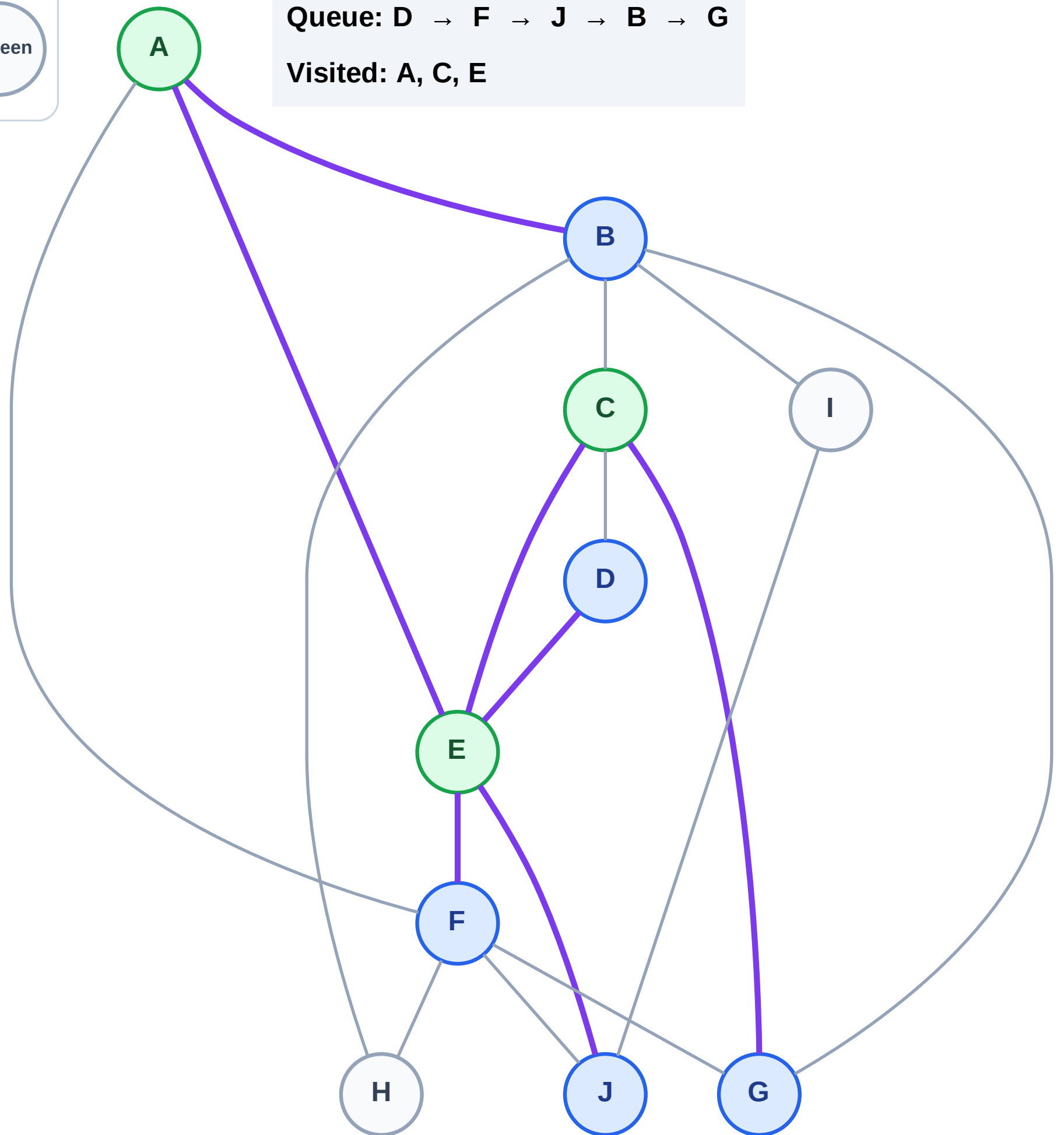
BFS Traversal

Step 7: Discover G from C



Queue: D → F → J → B → G

Visited: A, C, E



BFS Traversal

Step 8: Process node D

Legend

Complete

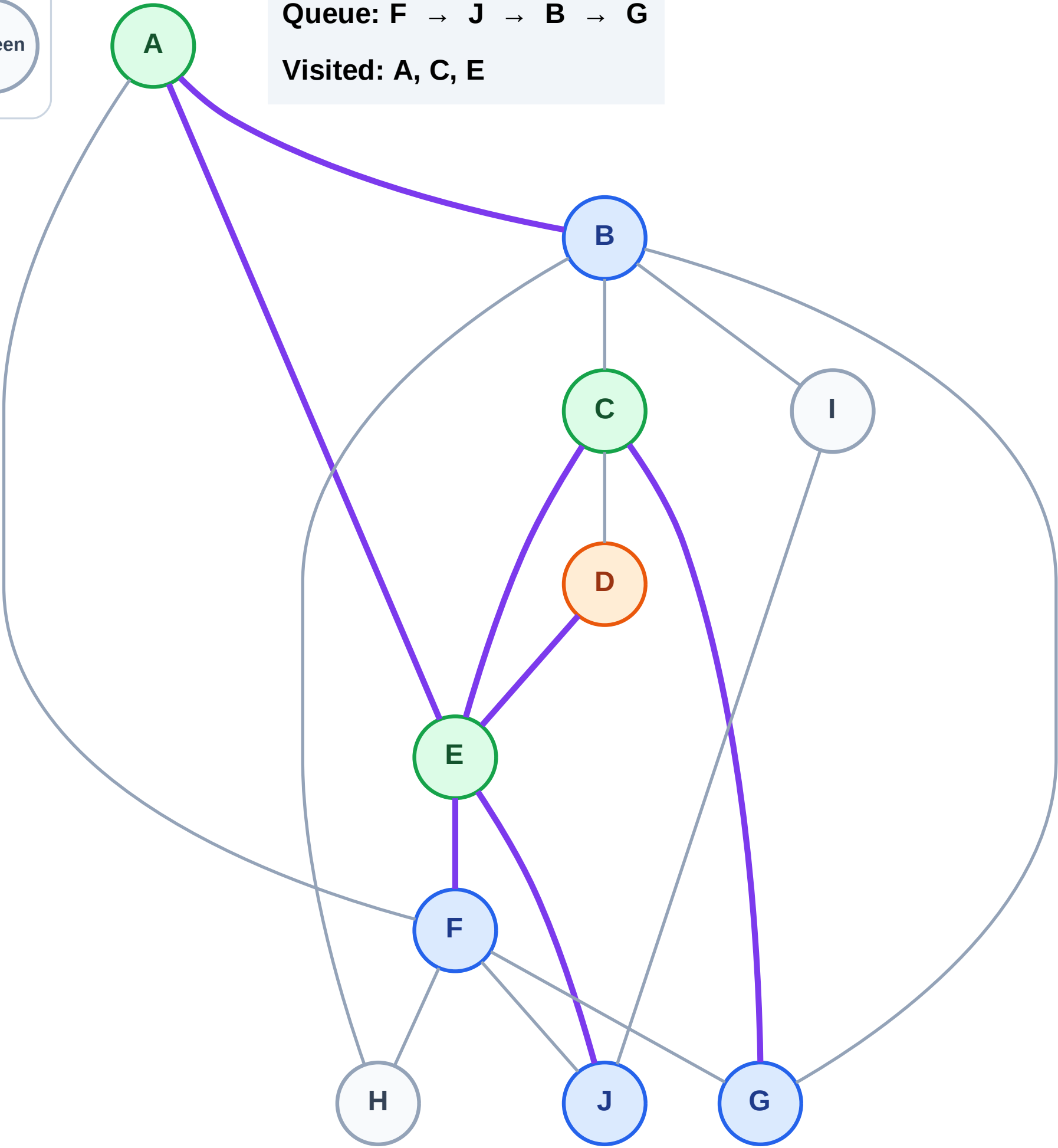
Processing

Discovered

Unseen

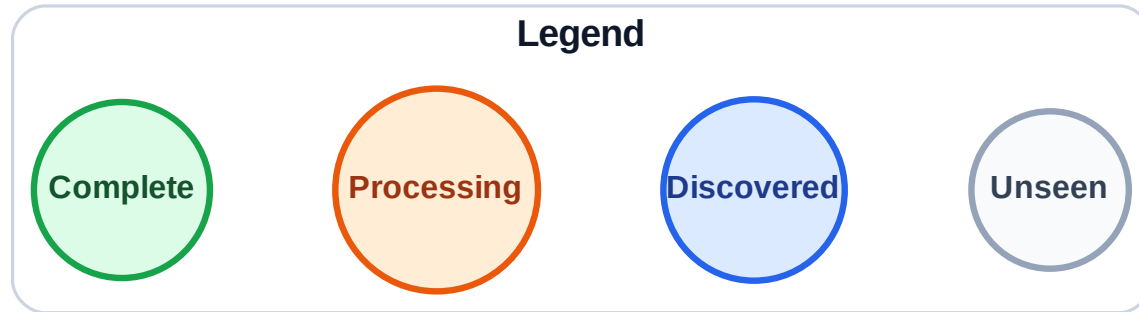
Queue: F → J → B → G

Visited: A, C, E



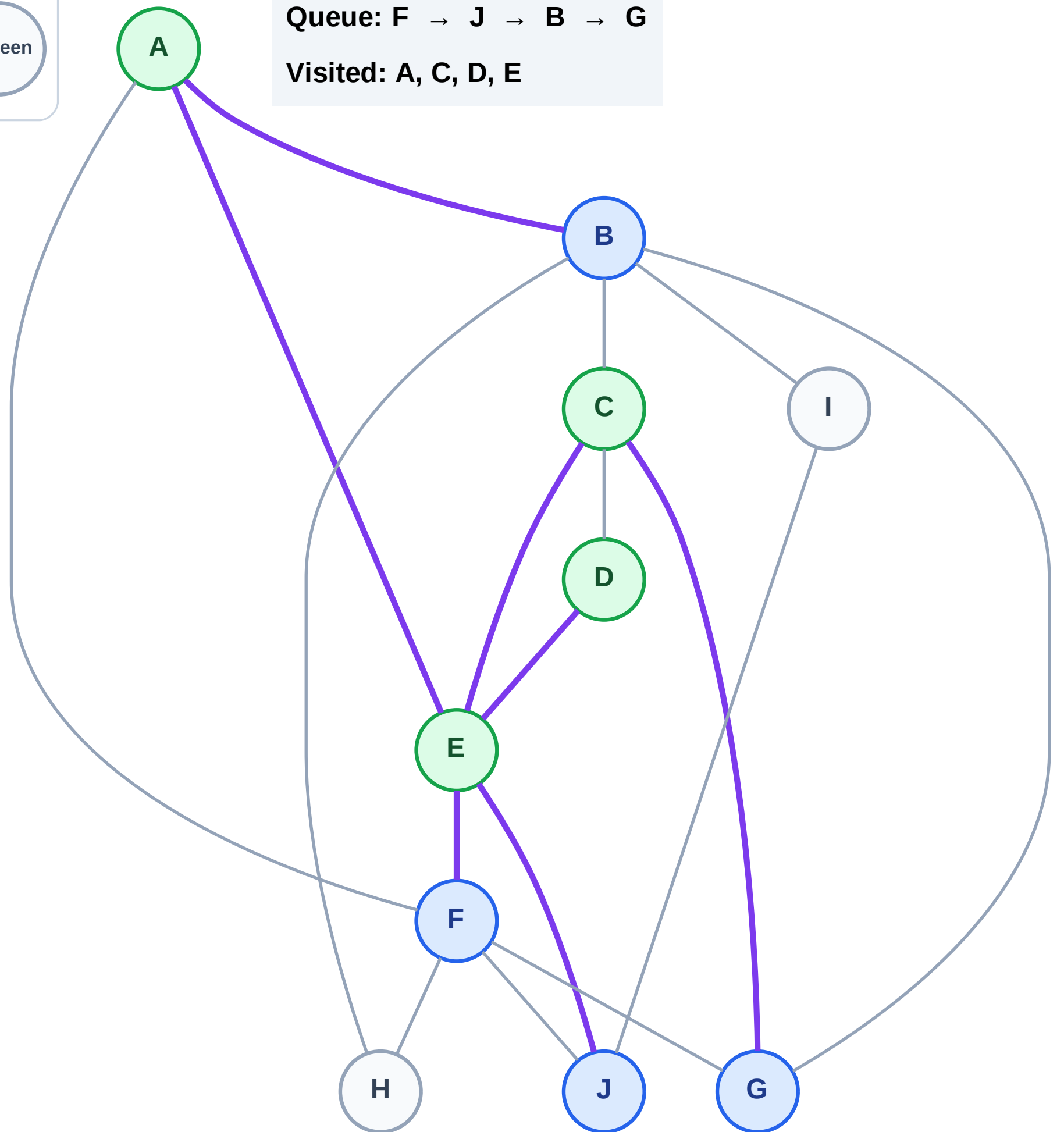
BFS Traversal

Step 9: No new neighbors from D



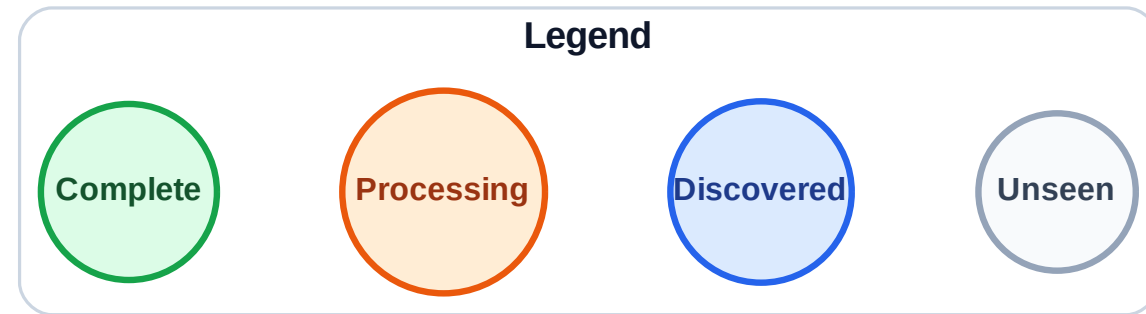
Queue: F → J → B → G

Visited: A, C, D, E



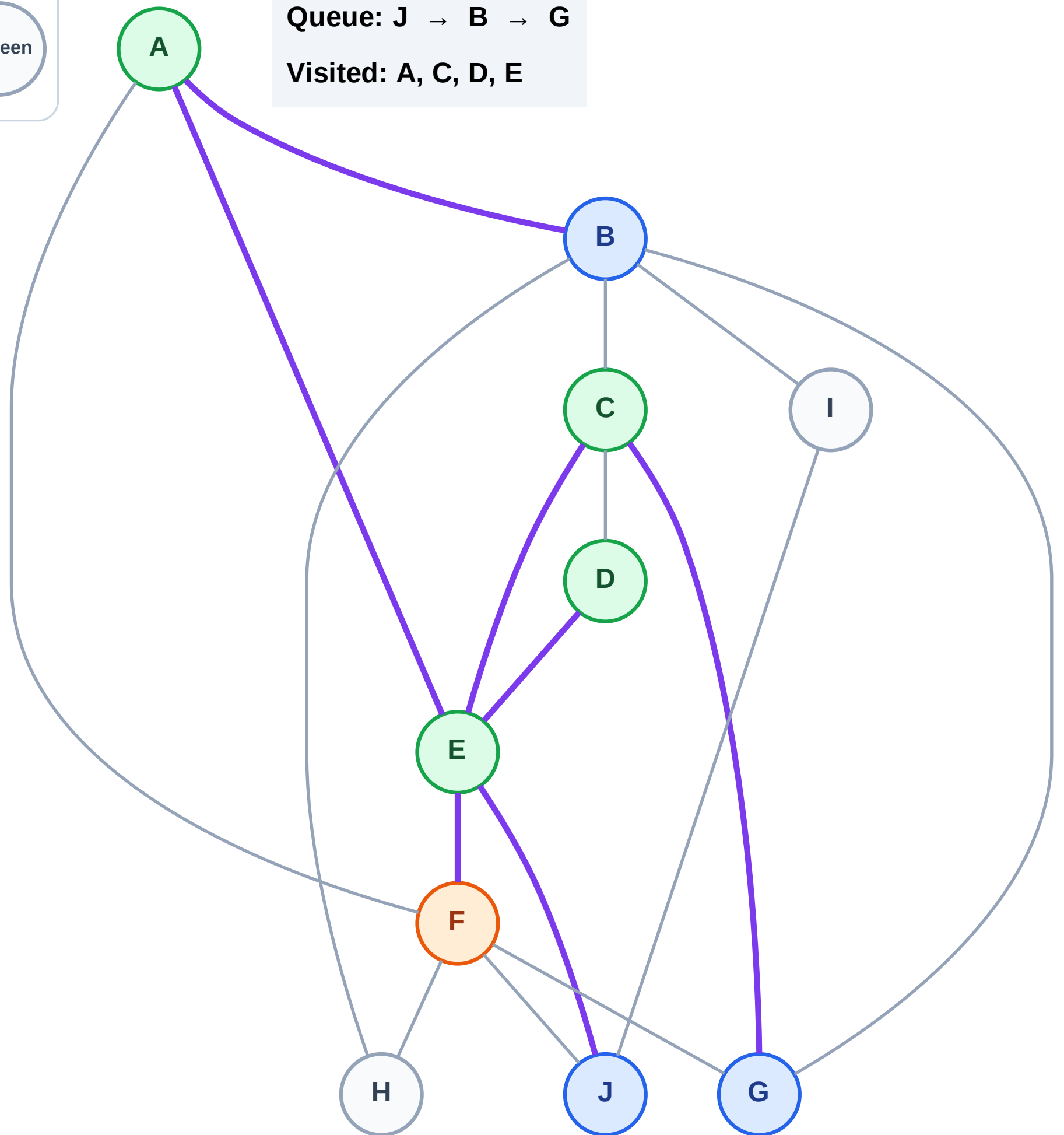
BFS Traversal

Step 10: Process node F



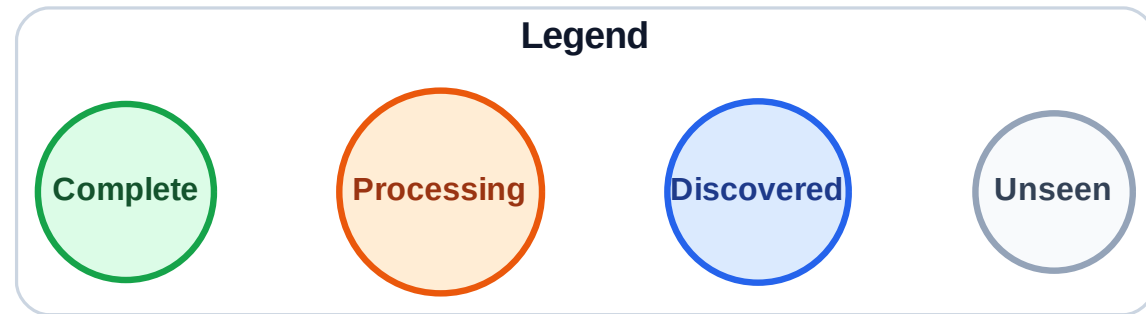
Queue: J → B → G

Visited: A, C, D, E

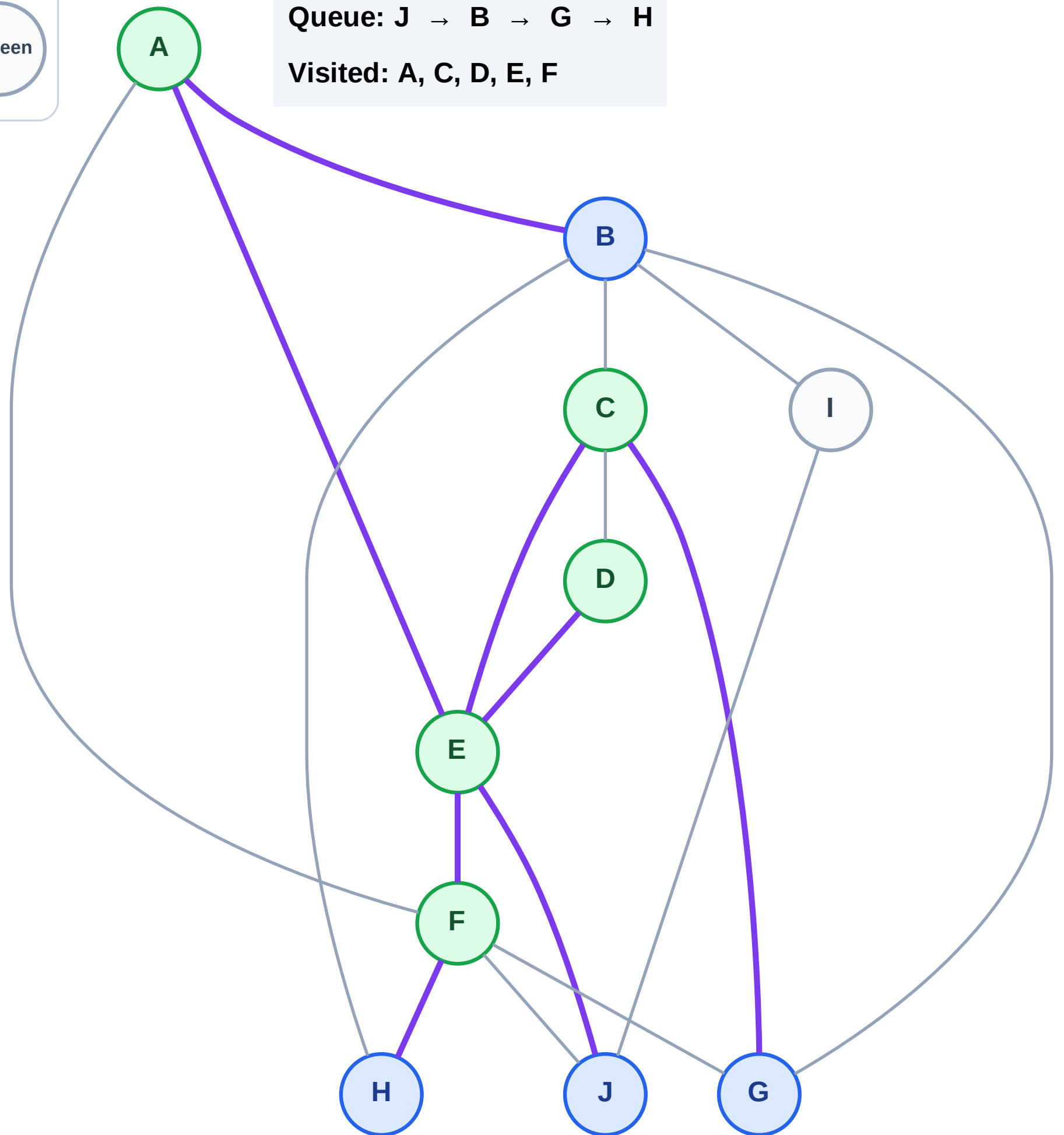


BFS Traversal

Step 11: Discover H from F



Queue: J → B → G → H
Visited: A, C, D, E, F



BFS Traversal

Step 12: Process node J

Legend

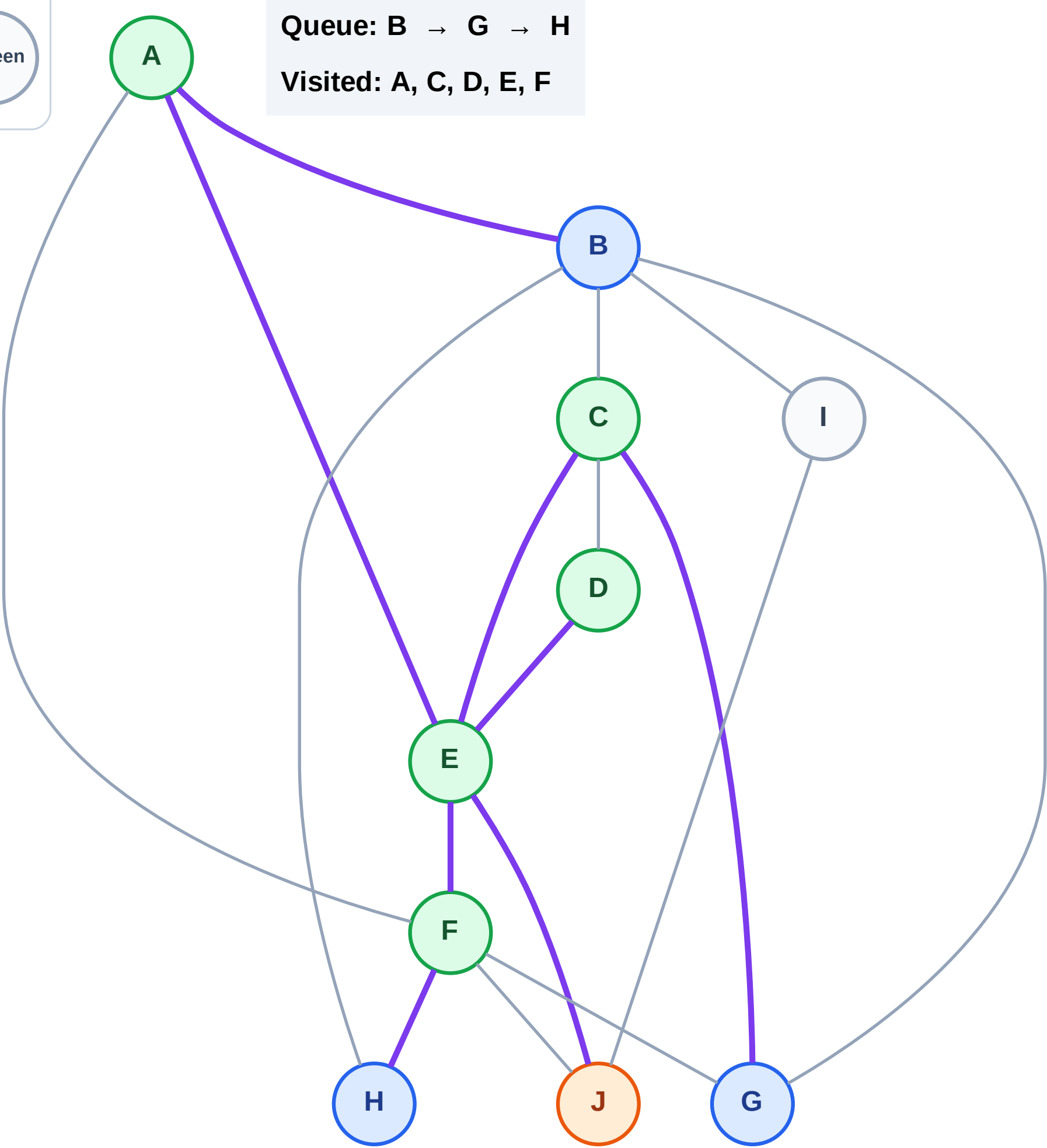
Complete

Processing

Discovered

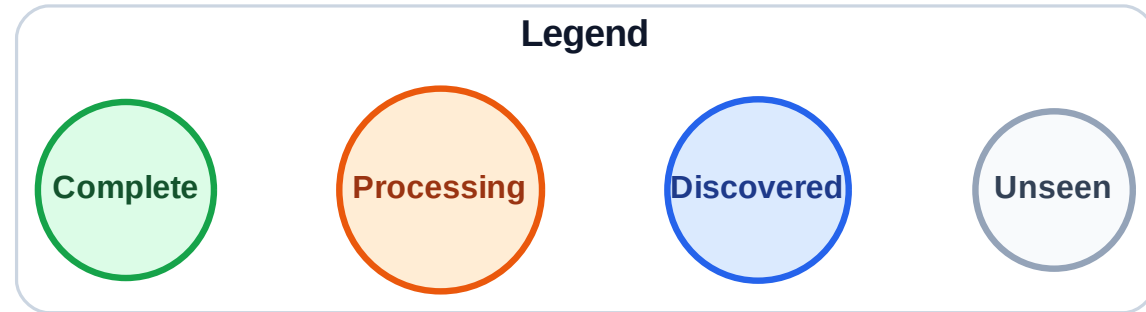
Unseen

Queue: B → G → H
Visited: A, C, D, E, F

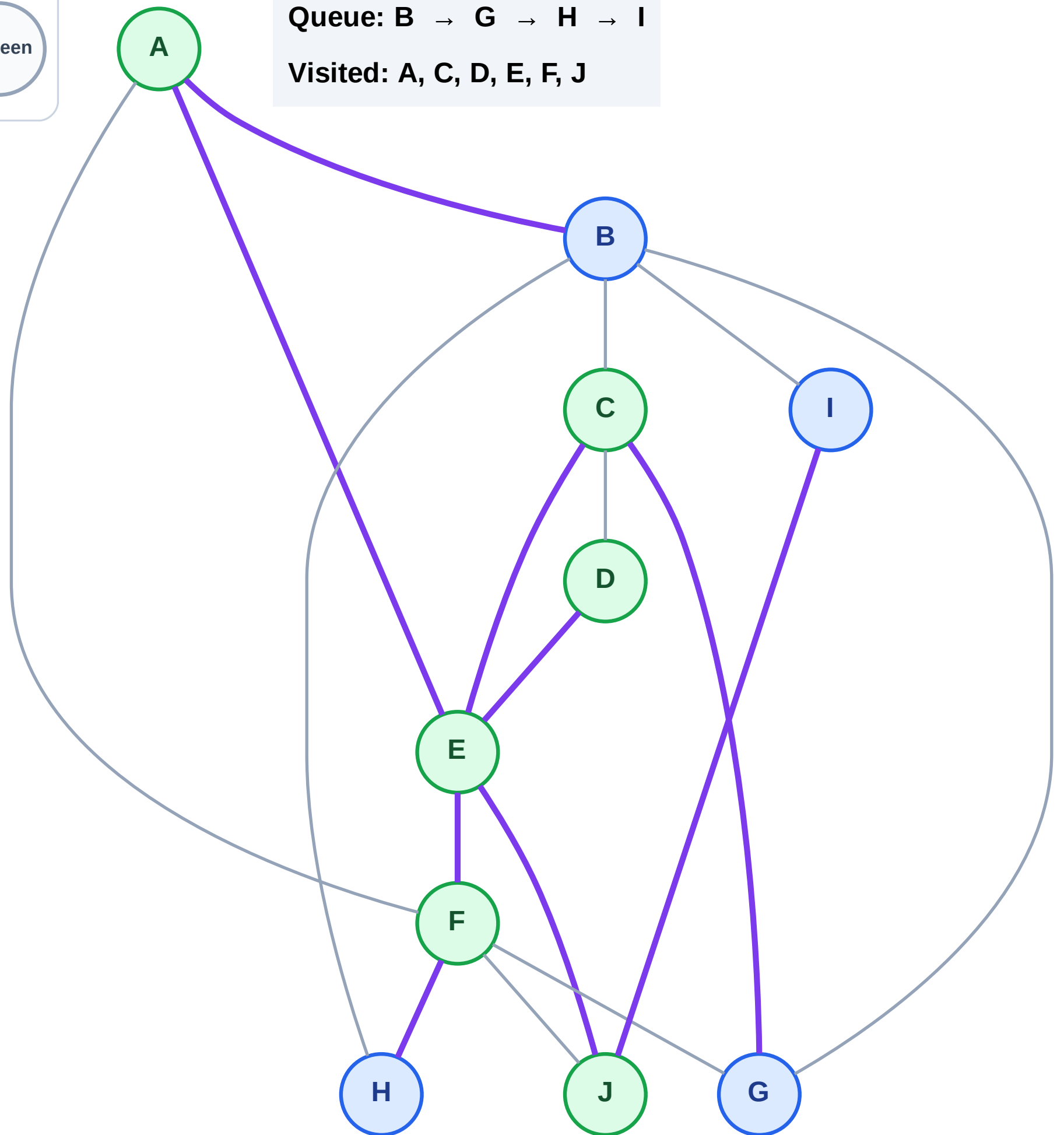


BFS Traversal

Step 13: Discover I from J



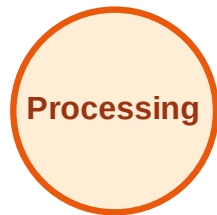
Queue: B → G → H → I
Visited: A, C, D, E, F, J



BFS Traversal

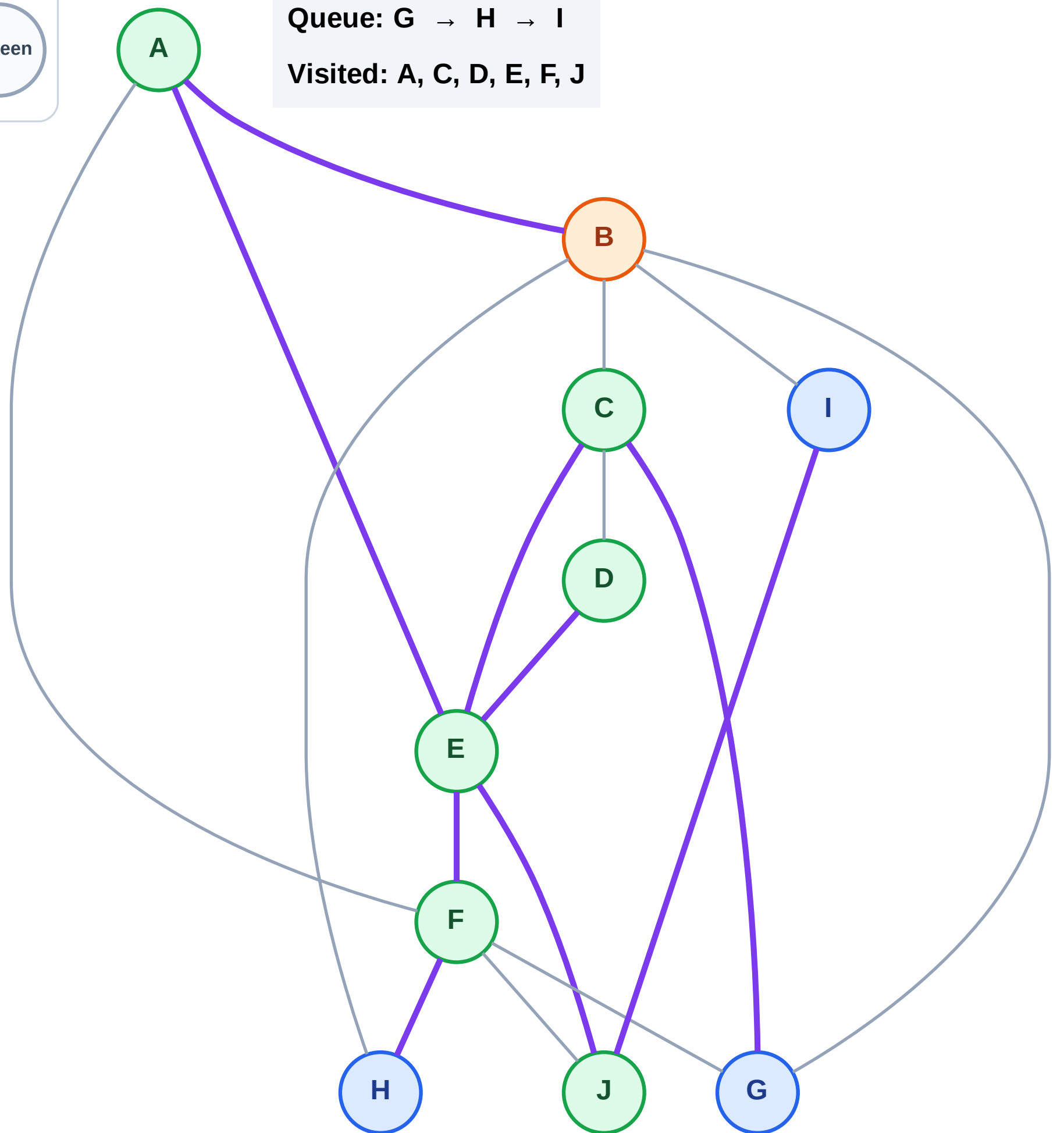
Step 14: Process node B

Legend



Queue: G → H → I

Visited: A, C, D, E, F, J



BFS Traversal

Step 15: No new neighbors from B

Legend

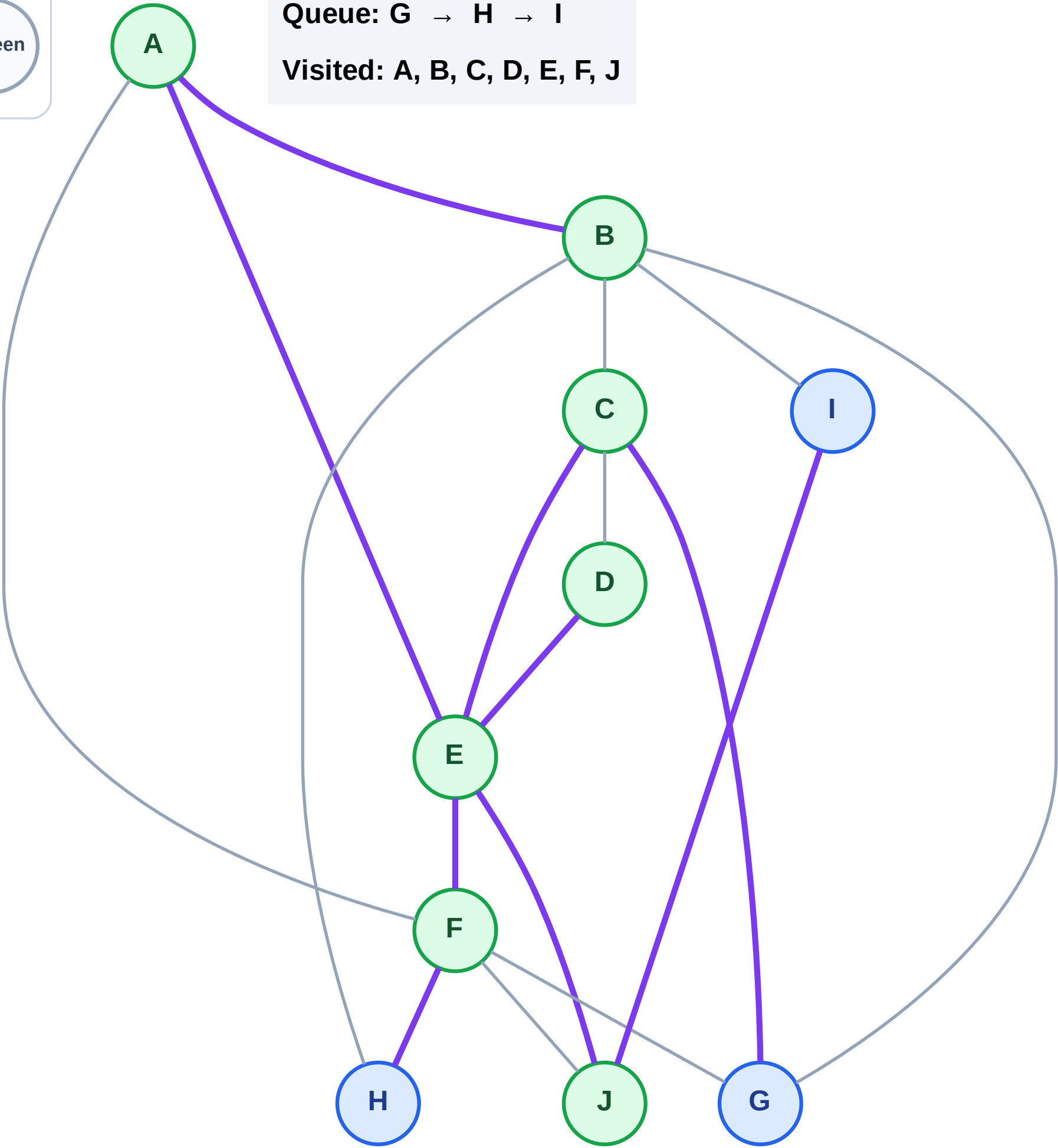
Complete

Processing

Discovered

Unseen

Queue: G → H → I
Visited: A, B, C, D, E, F, J



BFS Traversal

Step 16: Process node G

Legend



Complete



Processing



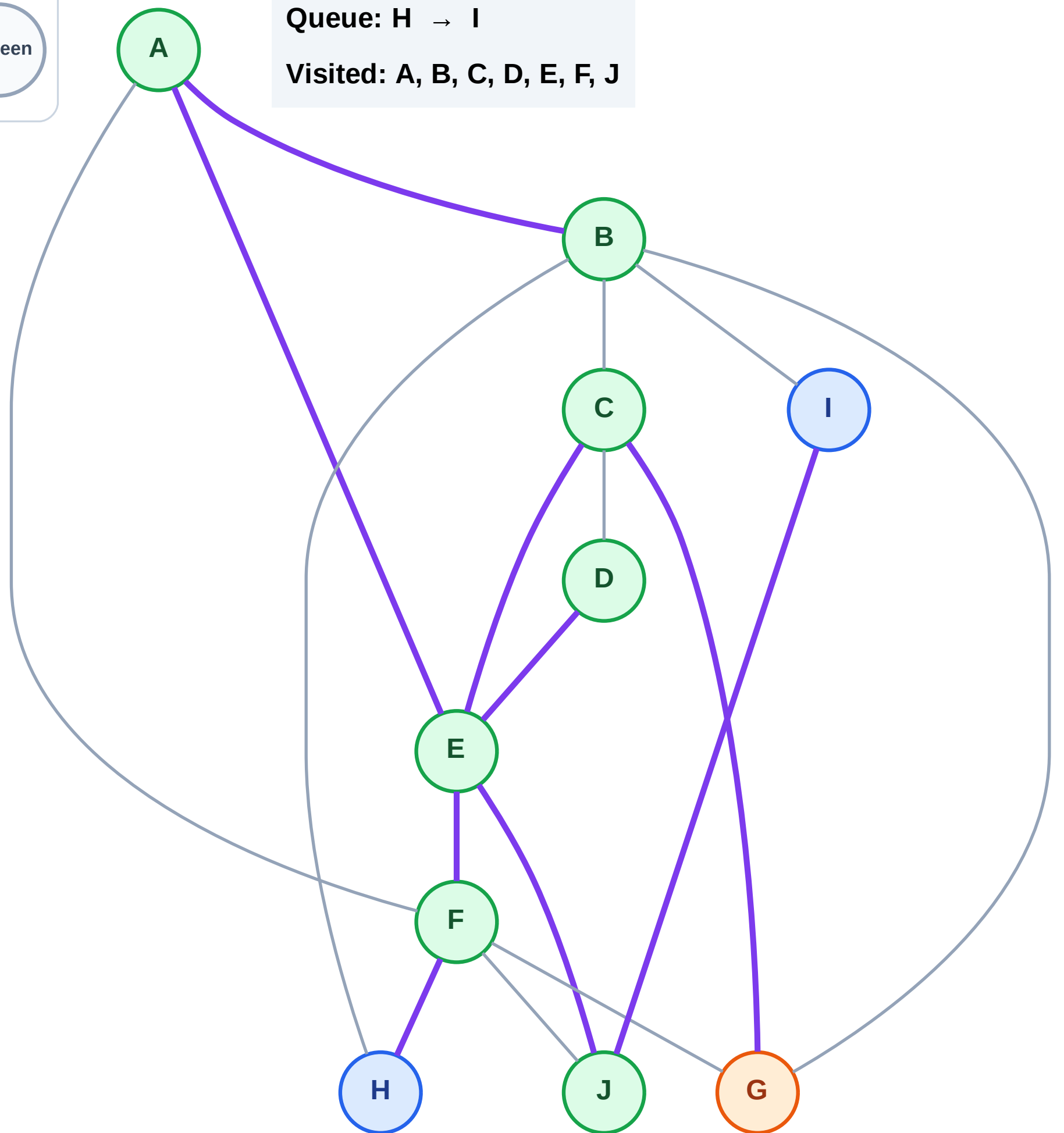
Discovered



Unseen

Queue: H → I

Visited: A, B, C, D, E, F, J



BFS Traversal

Step 17: No new neighbors from G

Legend

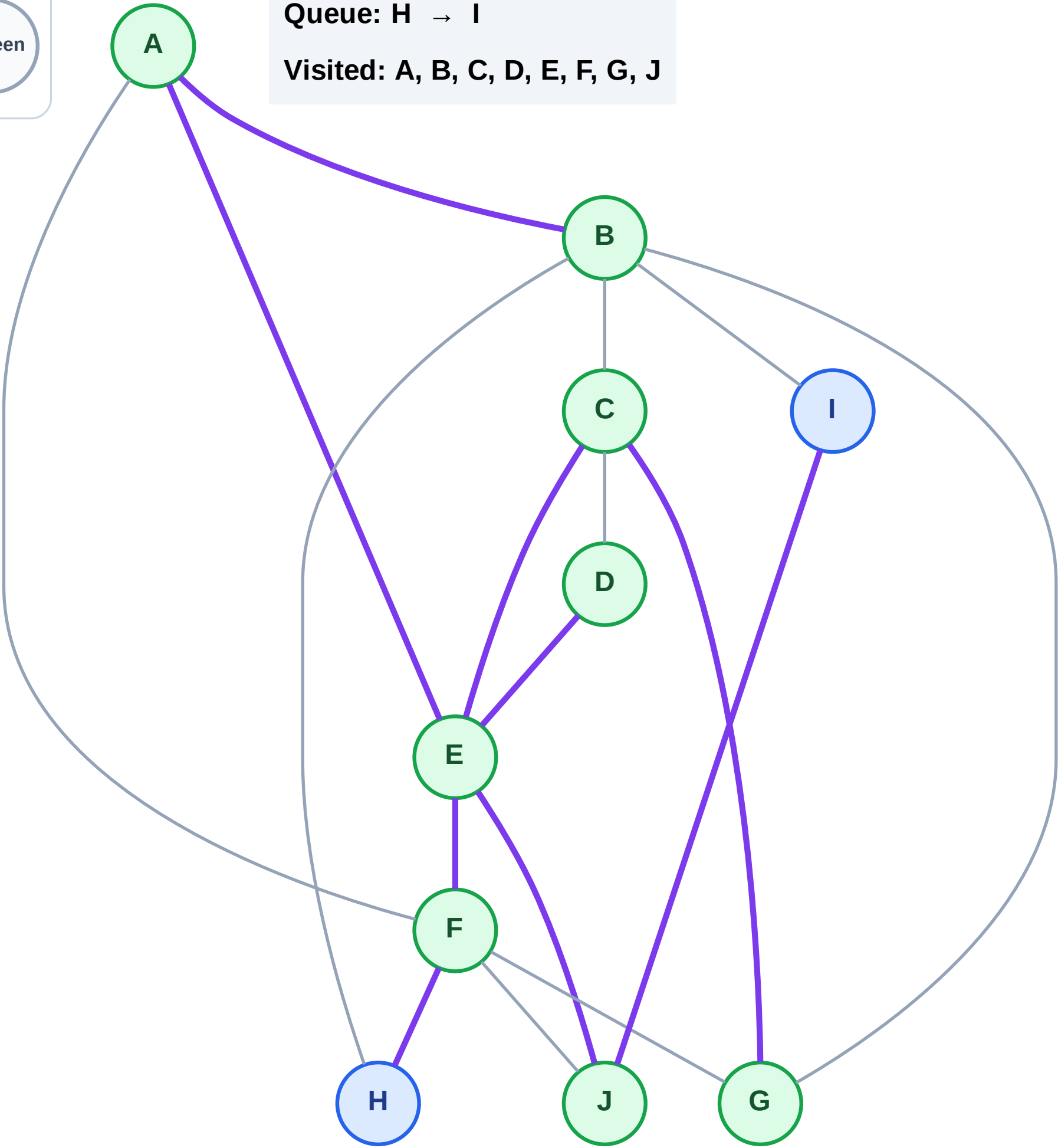
Complete

Processing

Discovered

Unseen

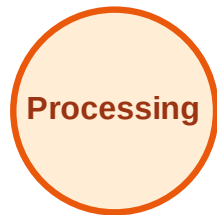
Queue: H → I
Visited: A, B, C, D, E, F, G, J



BFS Traversal

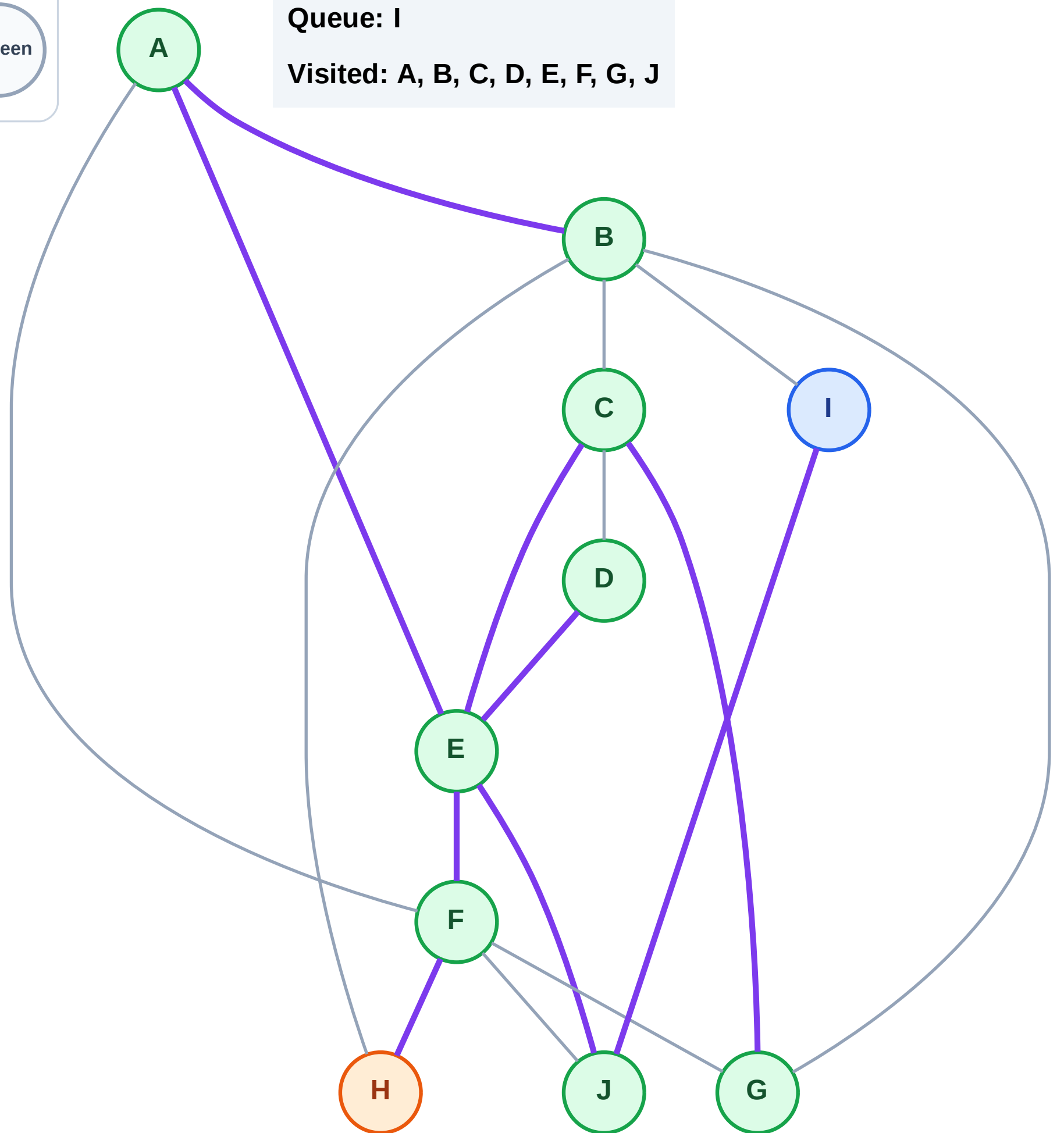
Step 18: Process node H

Legend



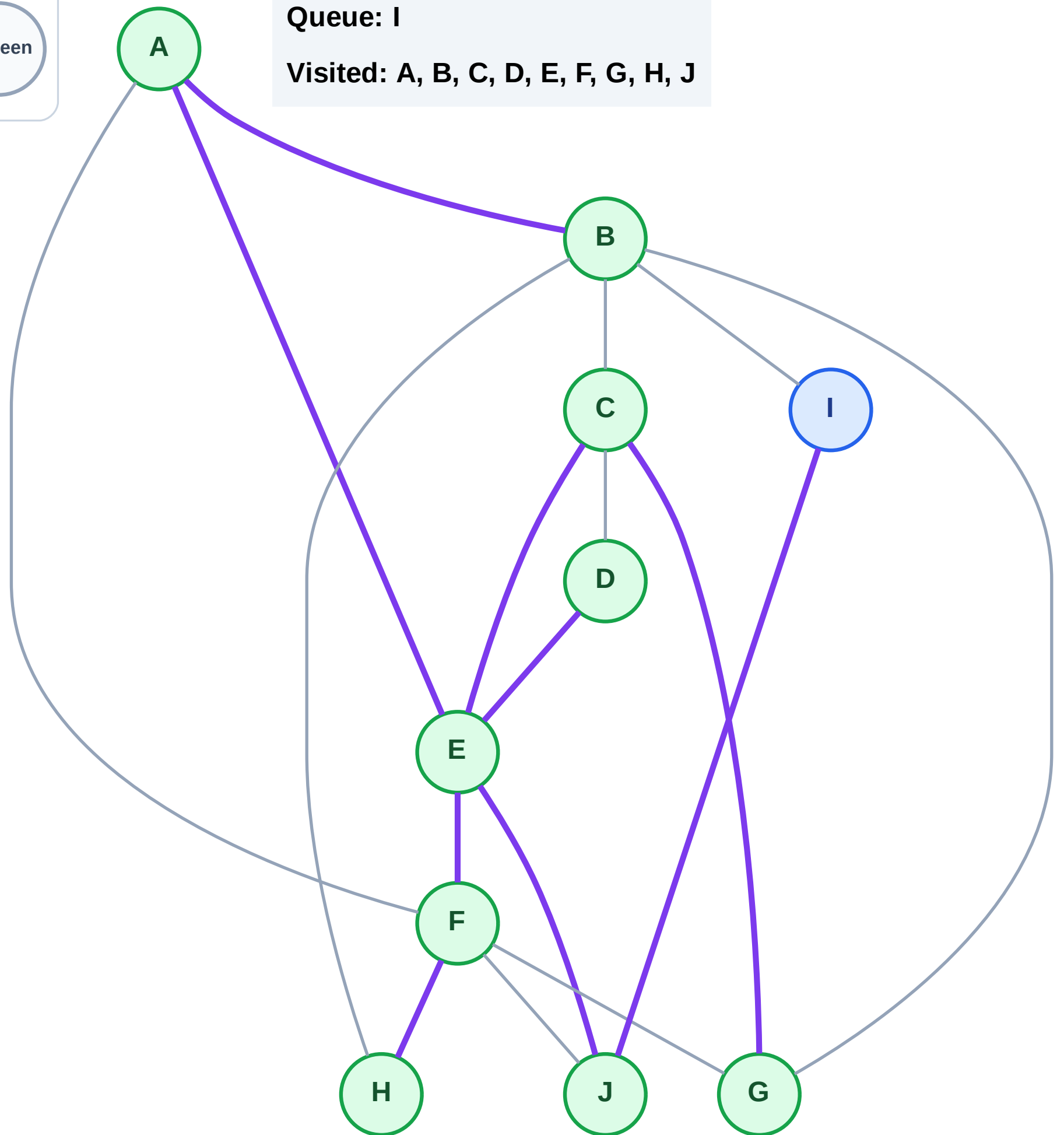
Queue: I

Visited: A, B, C, D, E, F, G, J



Step 19: No new neighbors from H

Step 19: No new neighbors from H



BFS Traversal

Step 20: Process node I

Legend

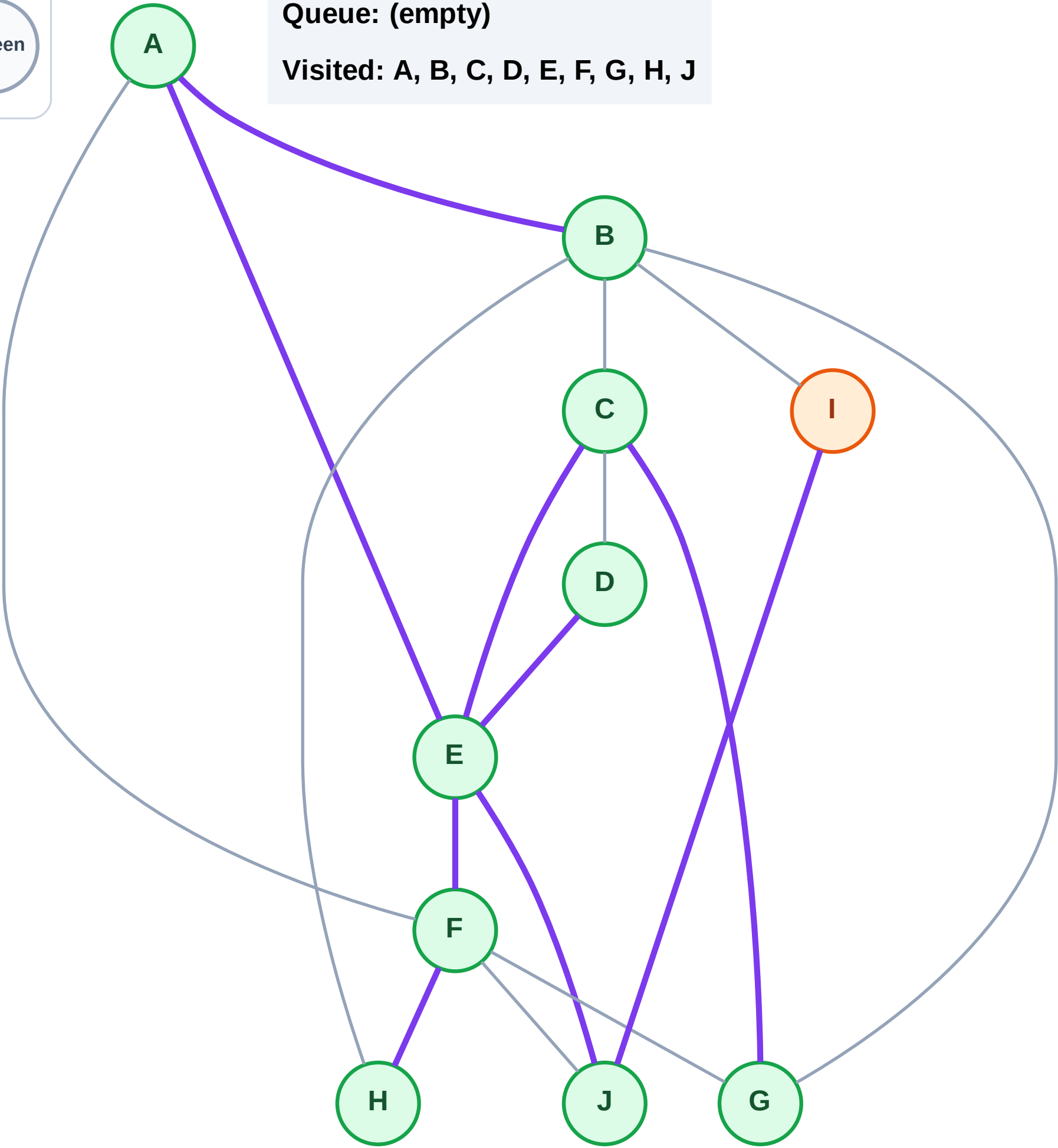
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

Step 21: No new neighbors from I

Legend

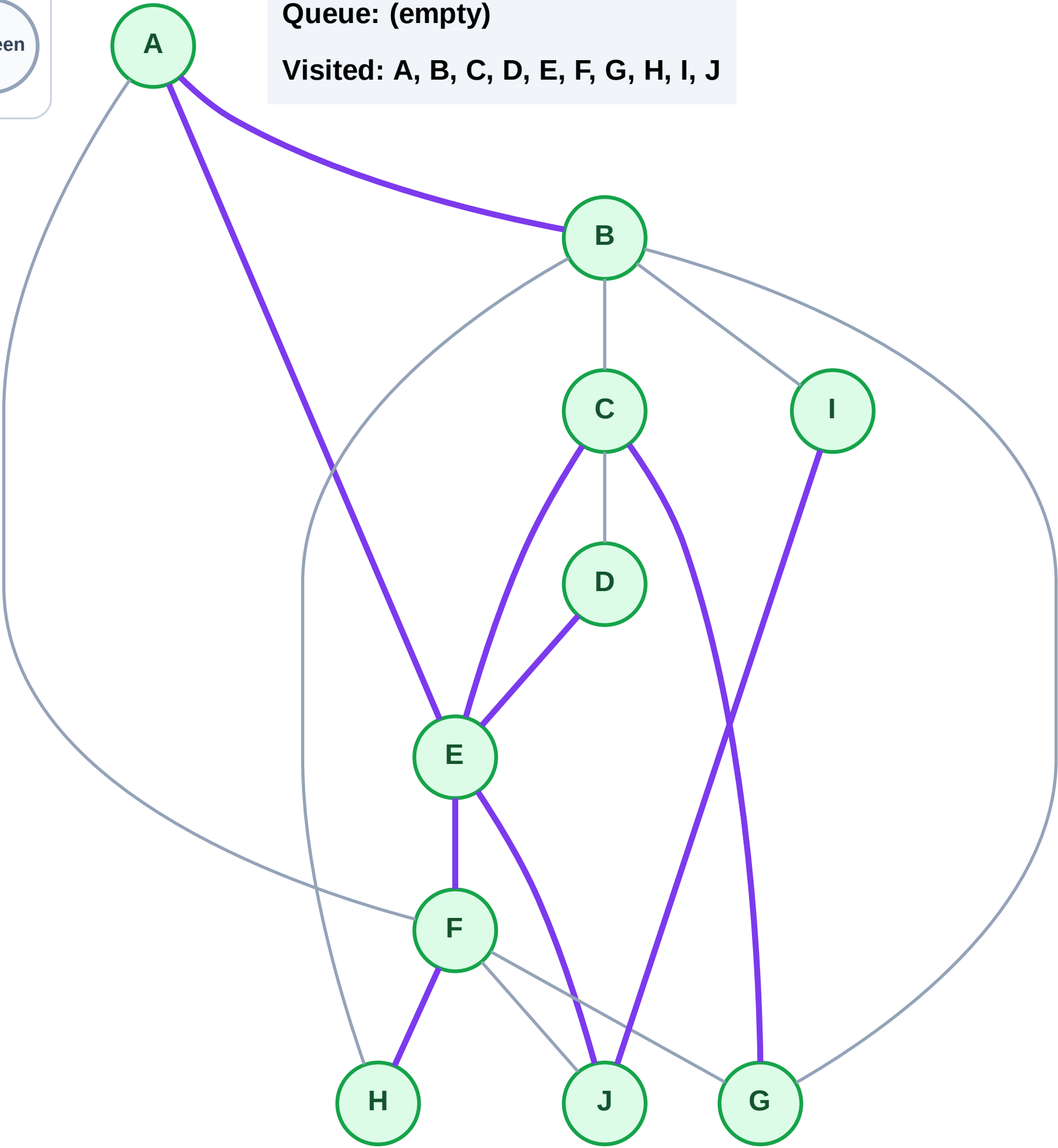
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at E

Legend

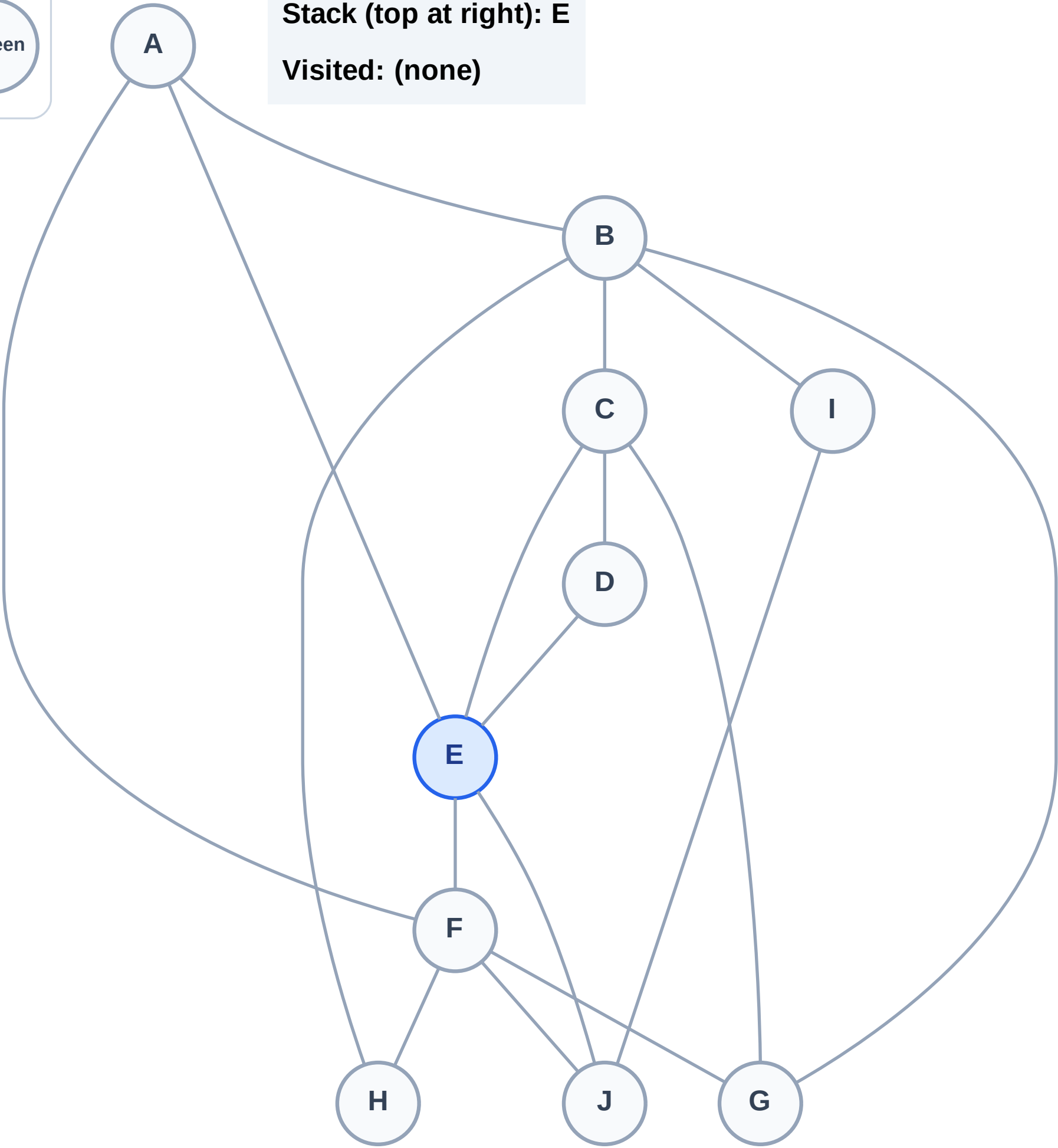
Complete

Processing

Discovered

Unseen

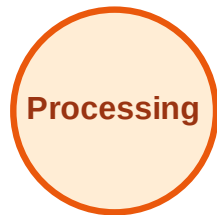
Stack (top at right): E
Visited: (none)



DFS Traversal

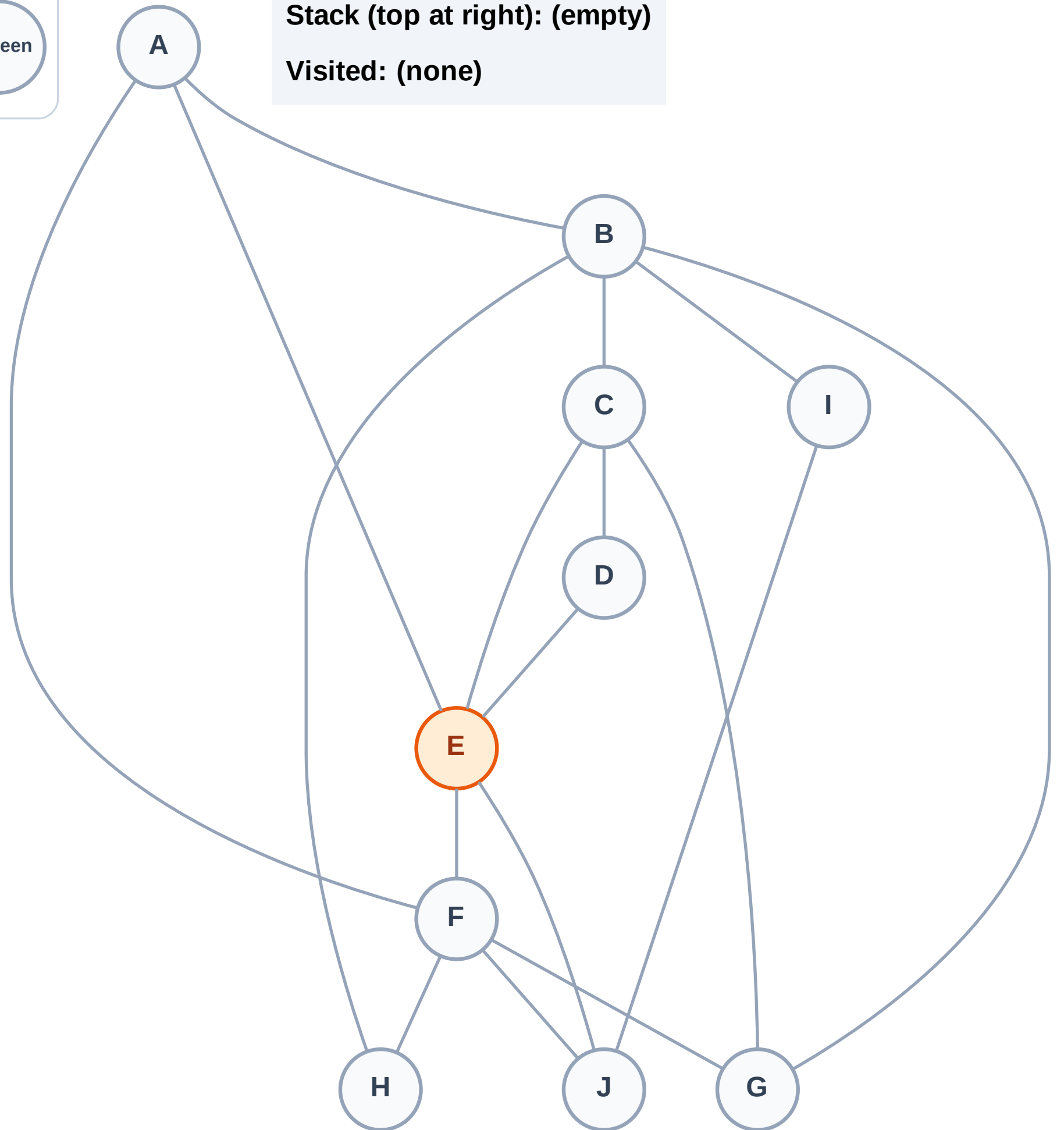
Step 2: Process node E

Legend



Stack (top at right): (empty)

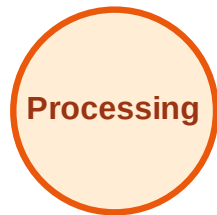
Visited: (none)



DFS Traversal

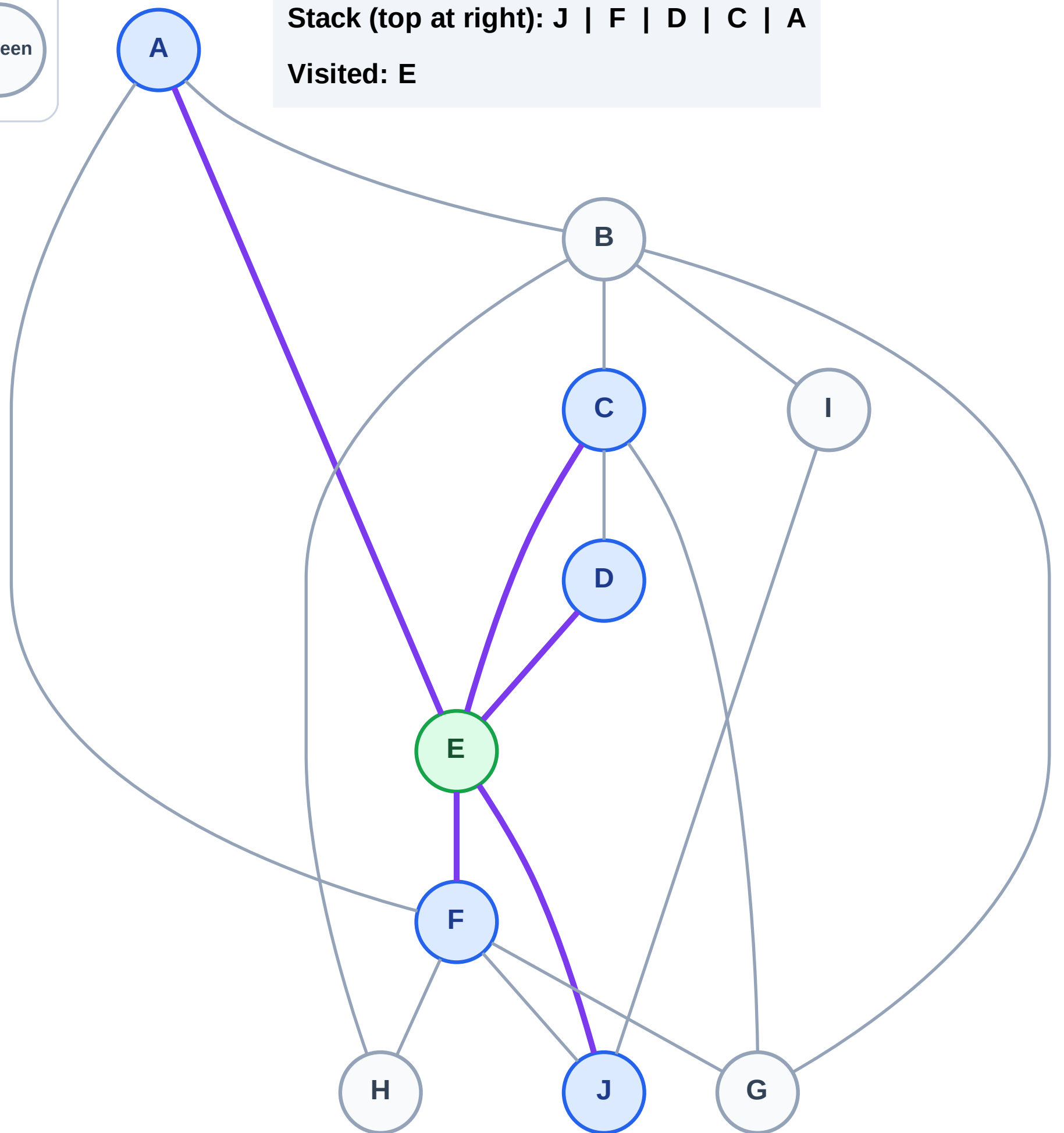
Step 3: Discover J, F, D, C, A from E

Legend



Stack (top at right): J | F | D | C | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

Complete

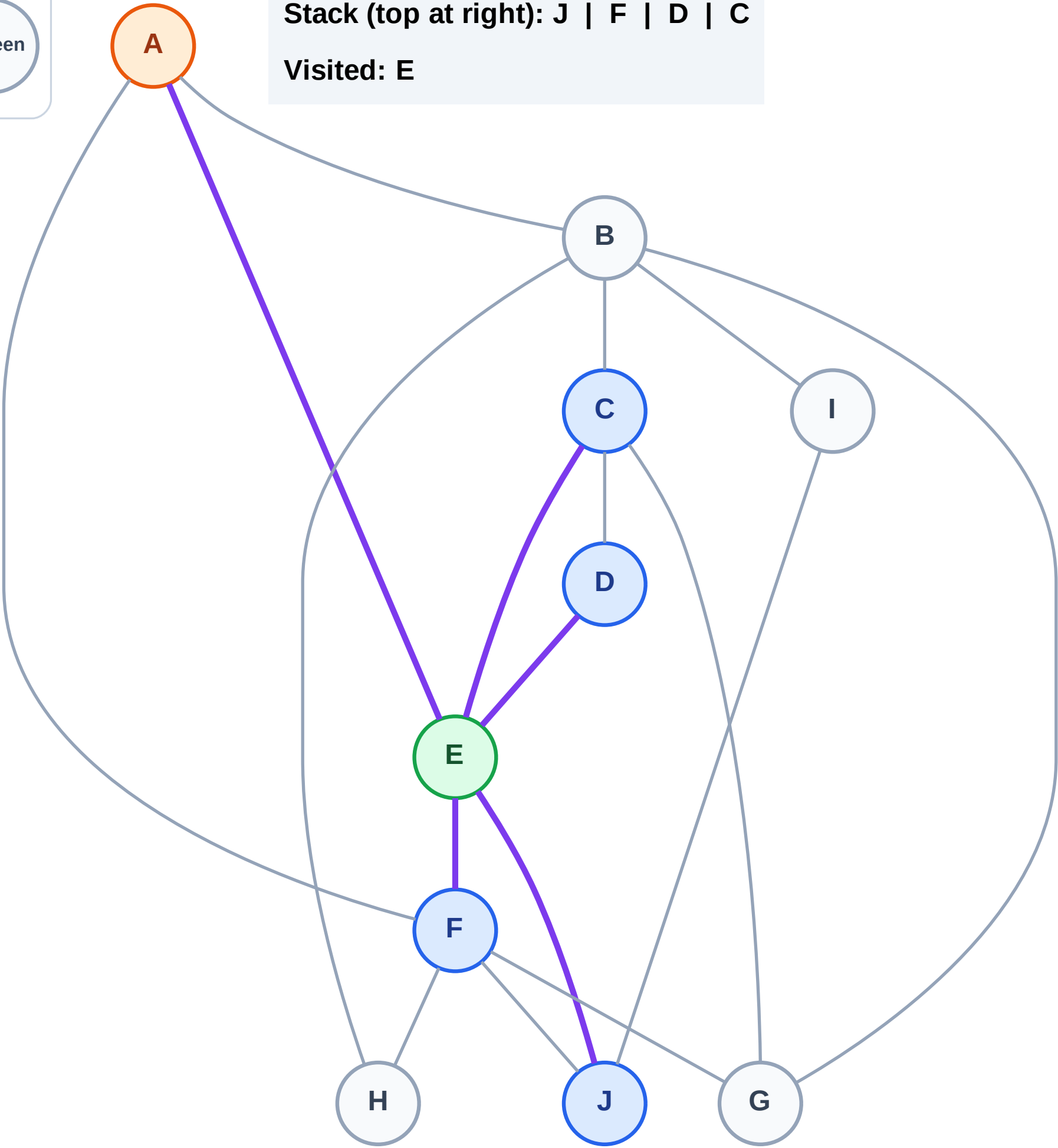
Processing

Discovered

Unseen

Stack (top at right): J | F | D | C

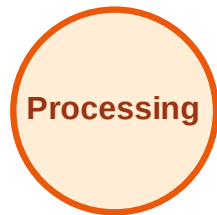
Visited: E



DFS Traversal

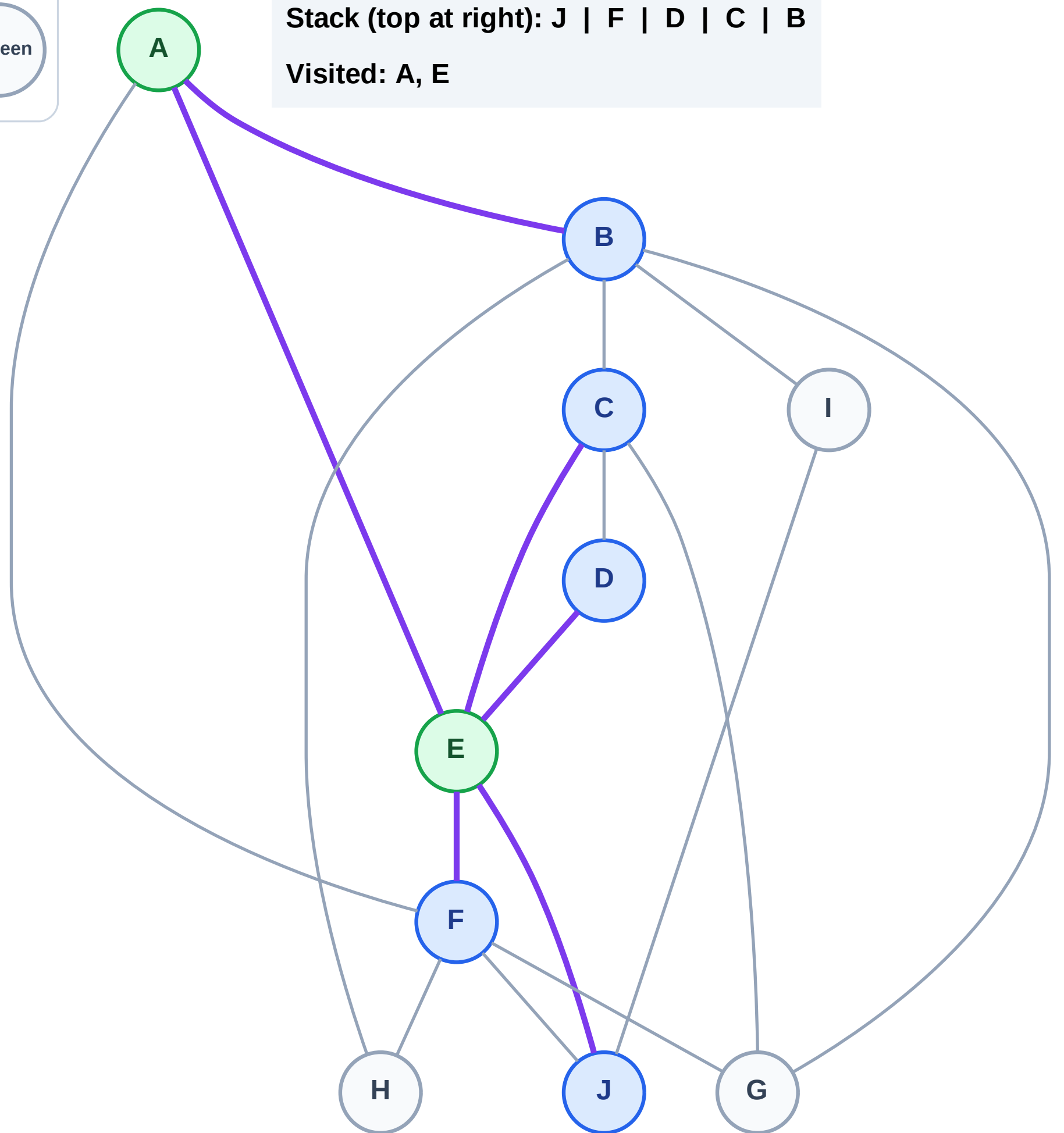
Step 5: Discover B from A

Legend



Stack (top at right): J | F | D | C | B

Visited: A, E



DFS Traversal

Step 6: Process node B

Legend

Complete

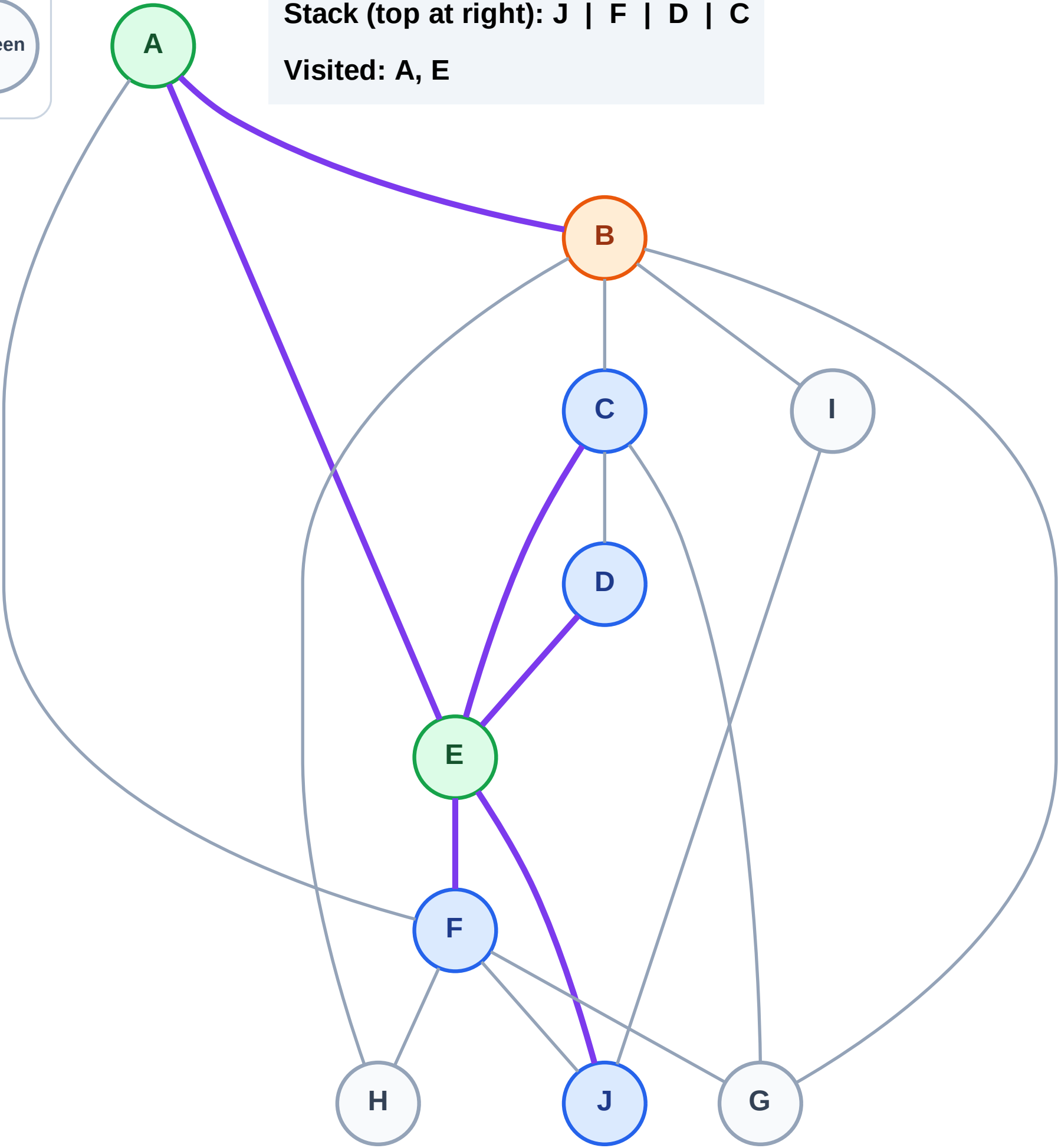
Processing

Discovered

Unseen

Stack (top at right): J | F | D | C

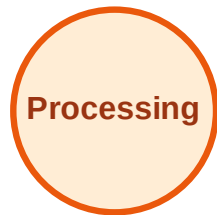
Visited: A, E



DFS Traversal

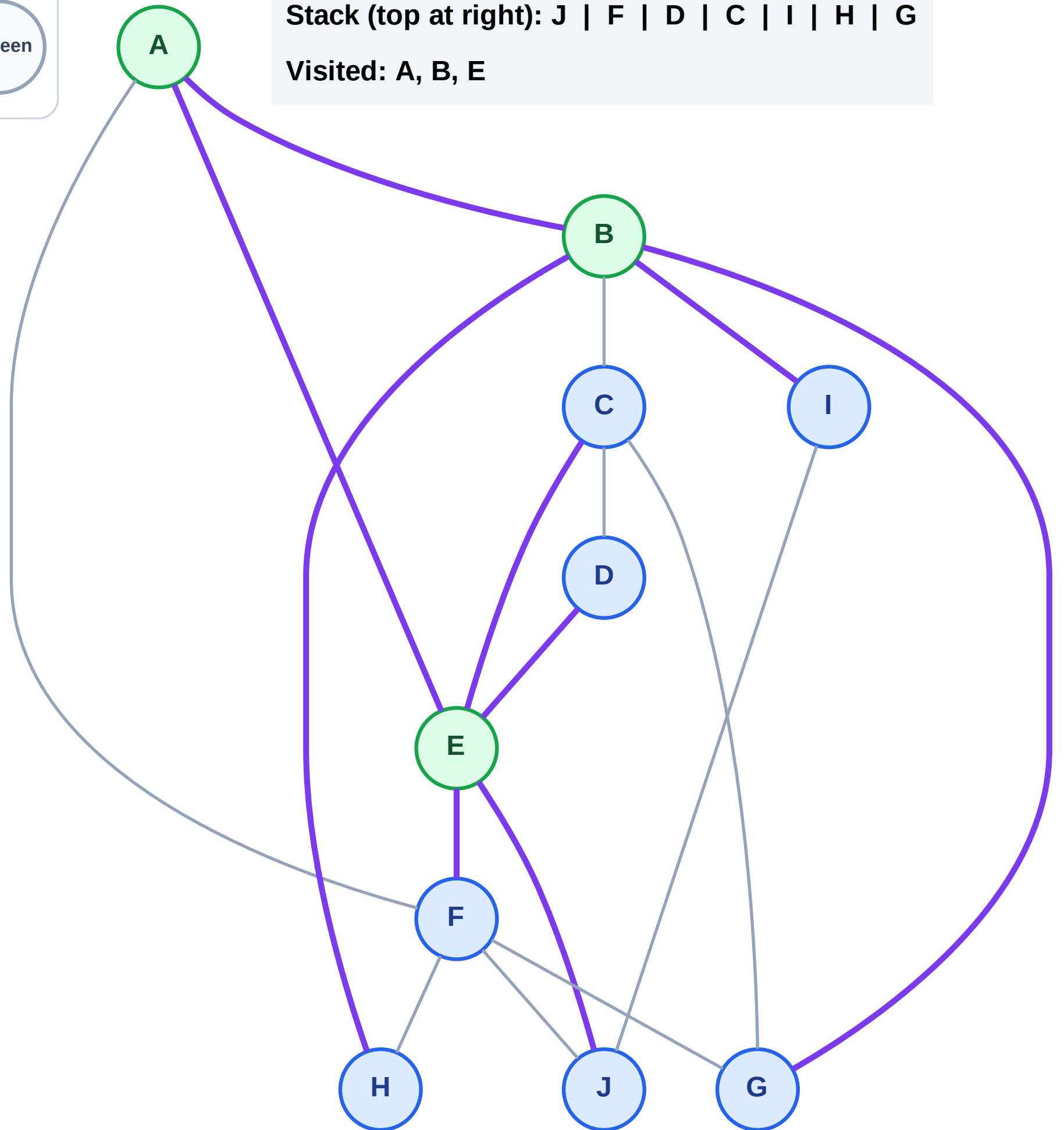
Step 7: Discover I, H, G from B

Legend



Stack (top at right): J | F | D | C | I | H | G

Visited: A, B, E



DFS Traversal

Step 8: Process node G

Legend

Complete

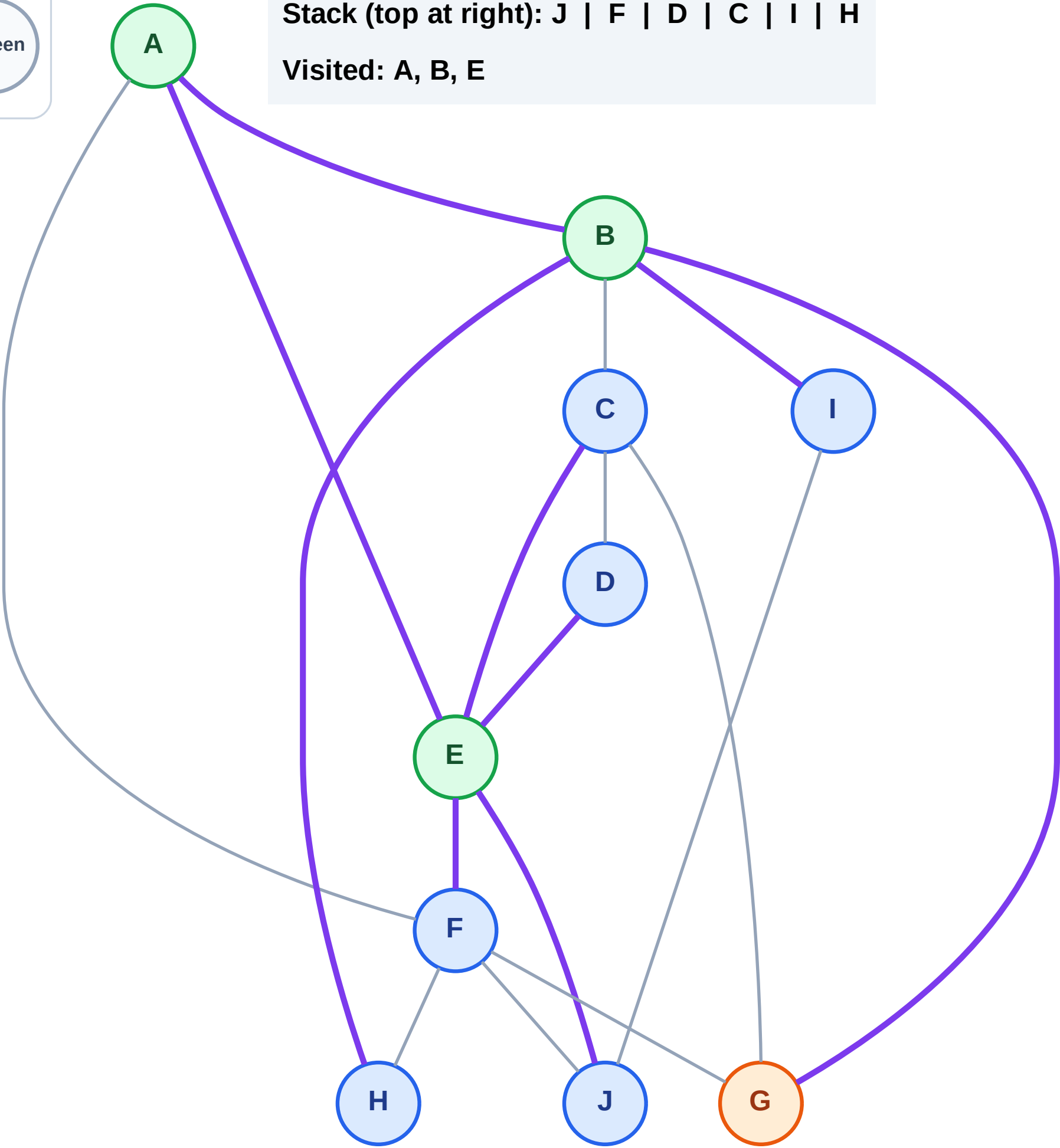
Processing

Discovered

Unseen

Stack (top at right): J | F | D | C | I | H

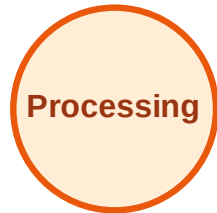
Visited: A, B, E



DFS Traversal

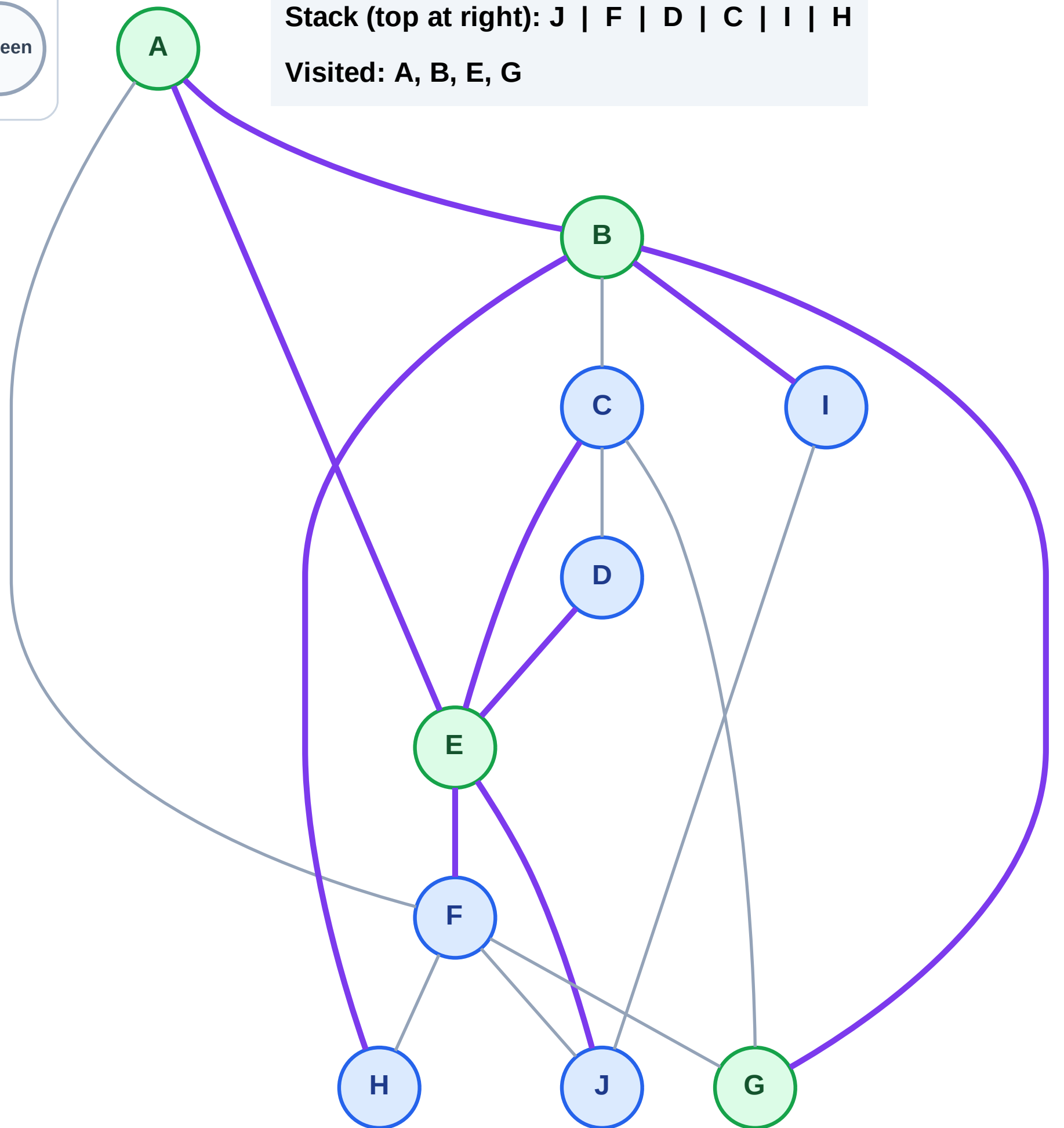
Step 9: No new neighbors from G

Legend



Stack (top at right): J | F | D | C | I | H

Visited: A, B, E, G



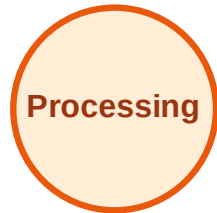
DFS Traversal

Step 10: Process node H

Legend



Complete



Processing



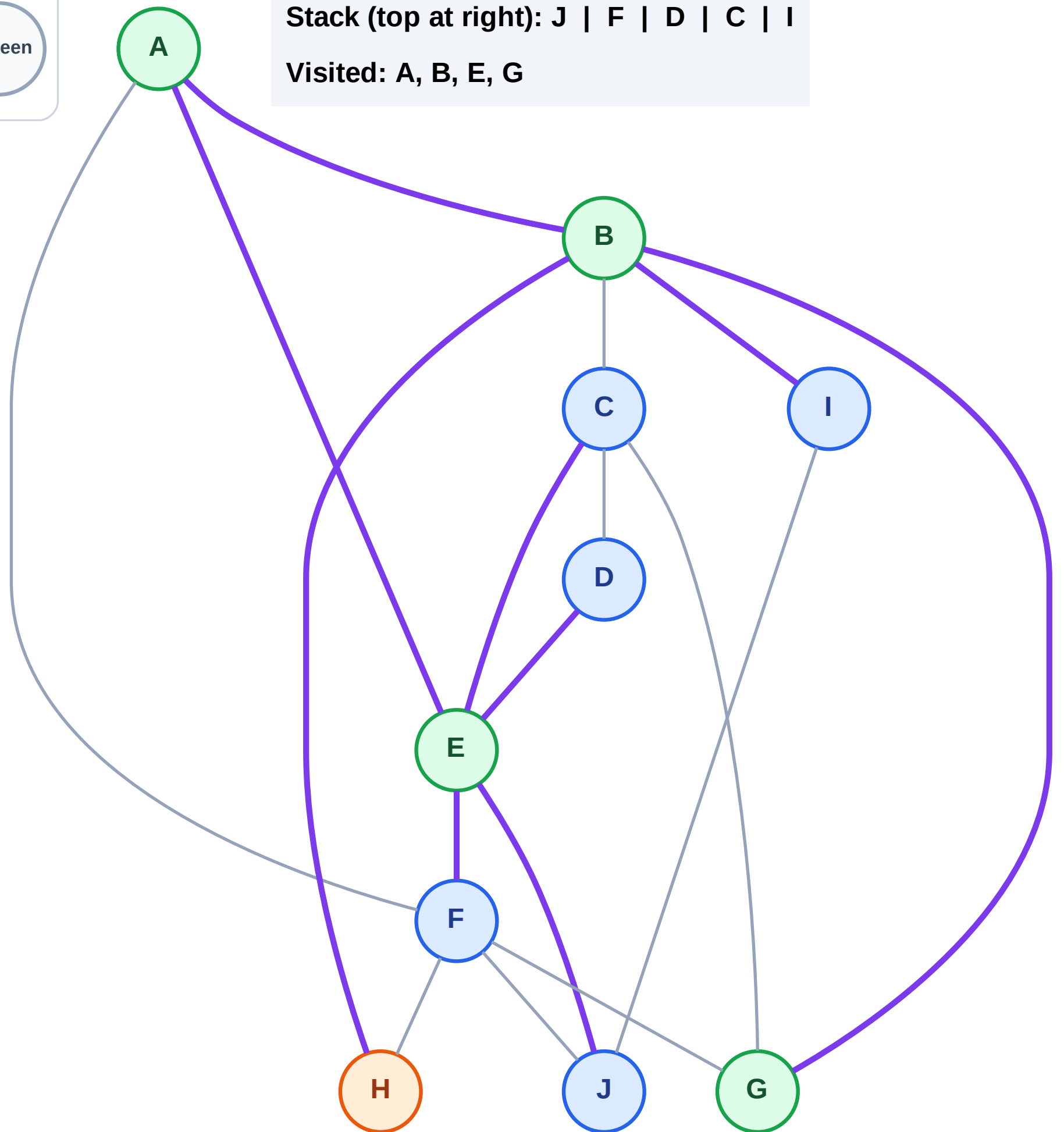
Discovered



Unseen

Stack (top at right): J | F | D | C | I

Visited: A, B, E, G



DFS Traversal

Step 11: No new neighbors from H

Legend

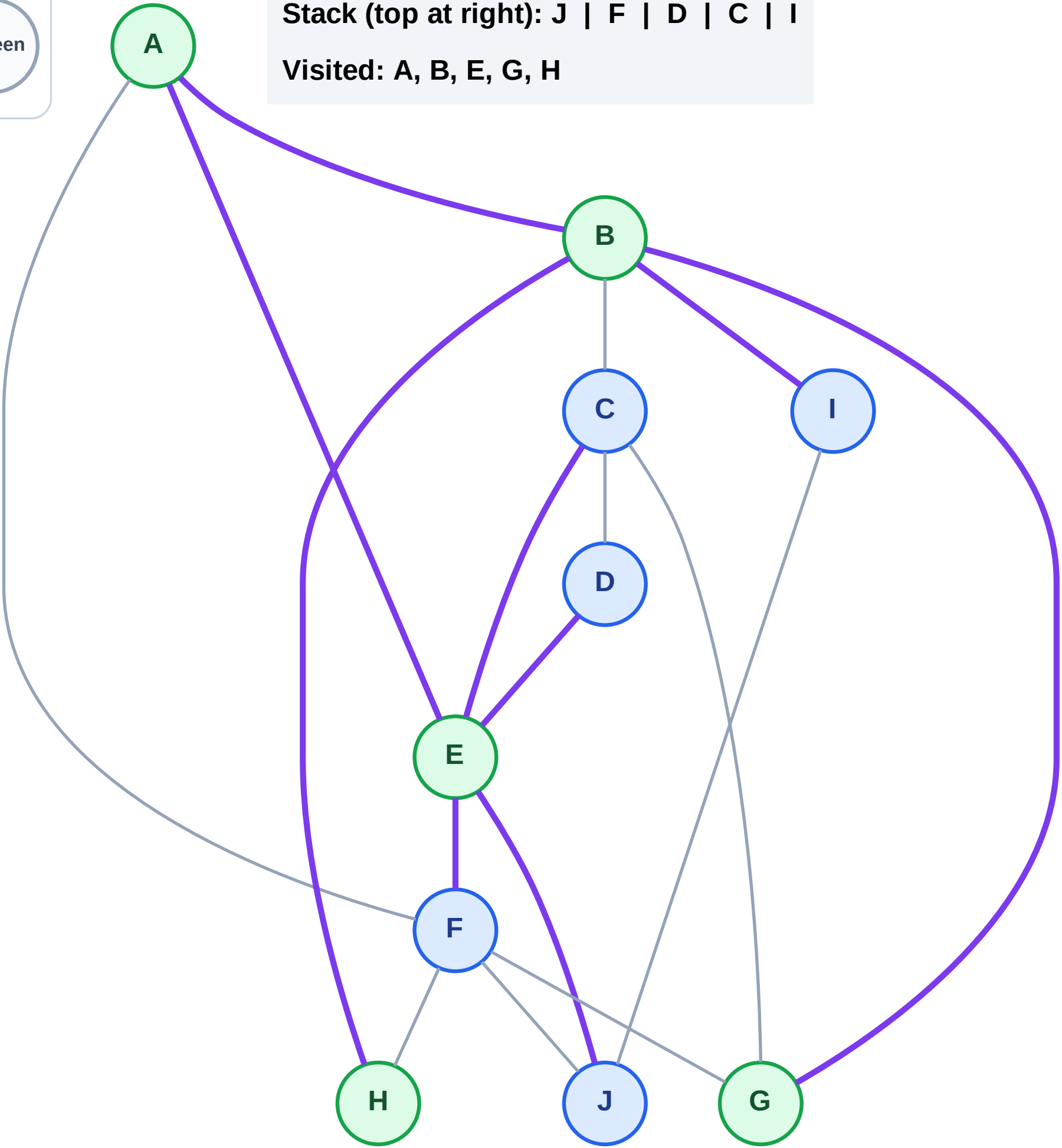
Complete

Processing

Discovered

Unseen

Stack (top at right): J | F | D | C | I
Visited: A, B, E, G, H



DFS Traversal

Step 12: Process node I

Legend

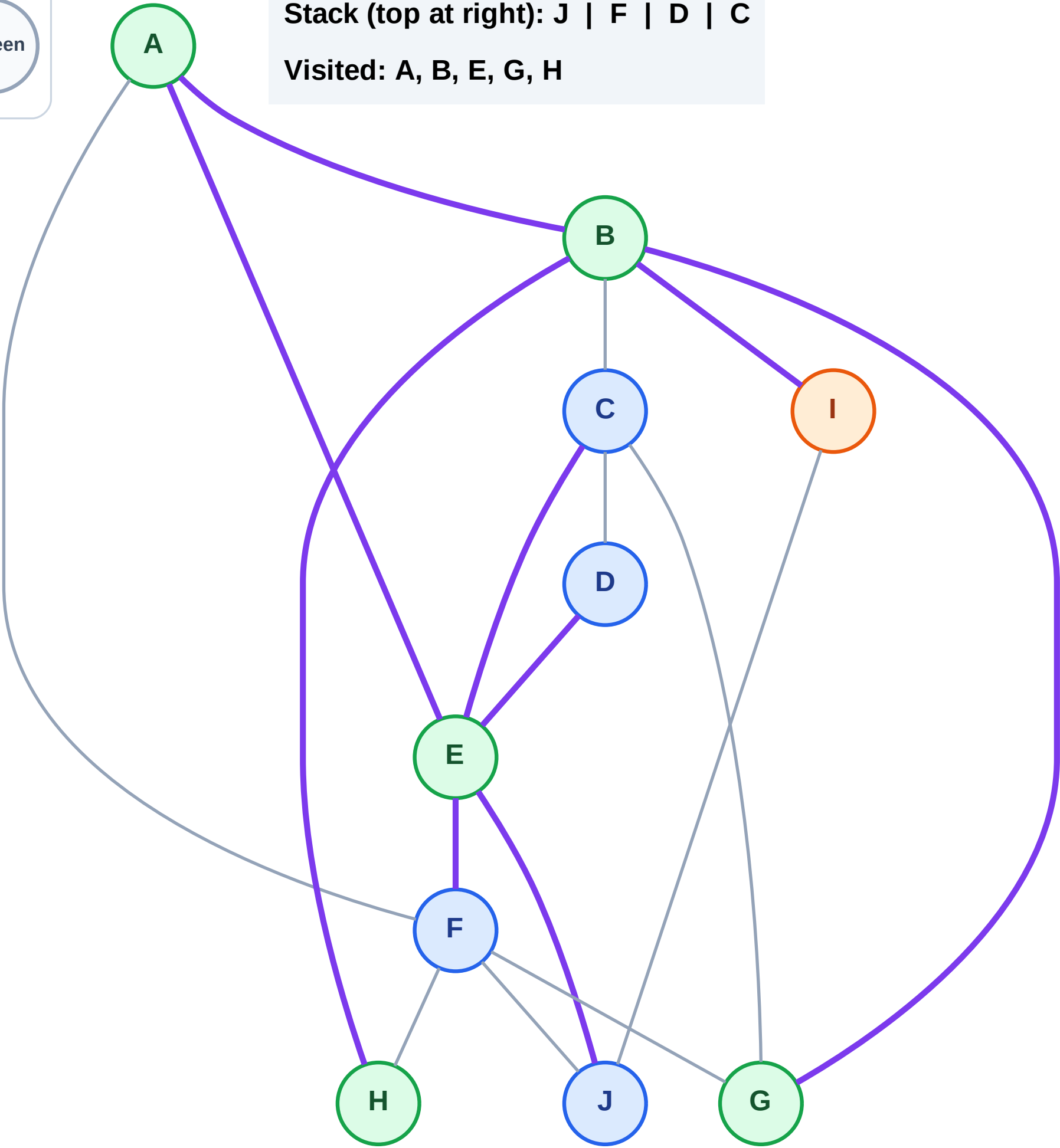
Complete

Processing

Discovered

Unseen

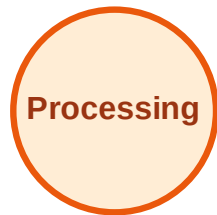
Stack (top at right): J | F | D | C
Visited: A, B, E, G, H



DFS Traversal

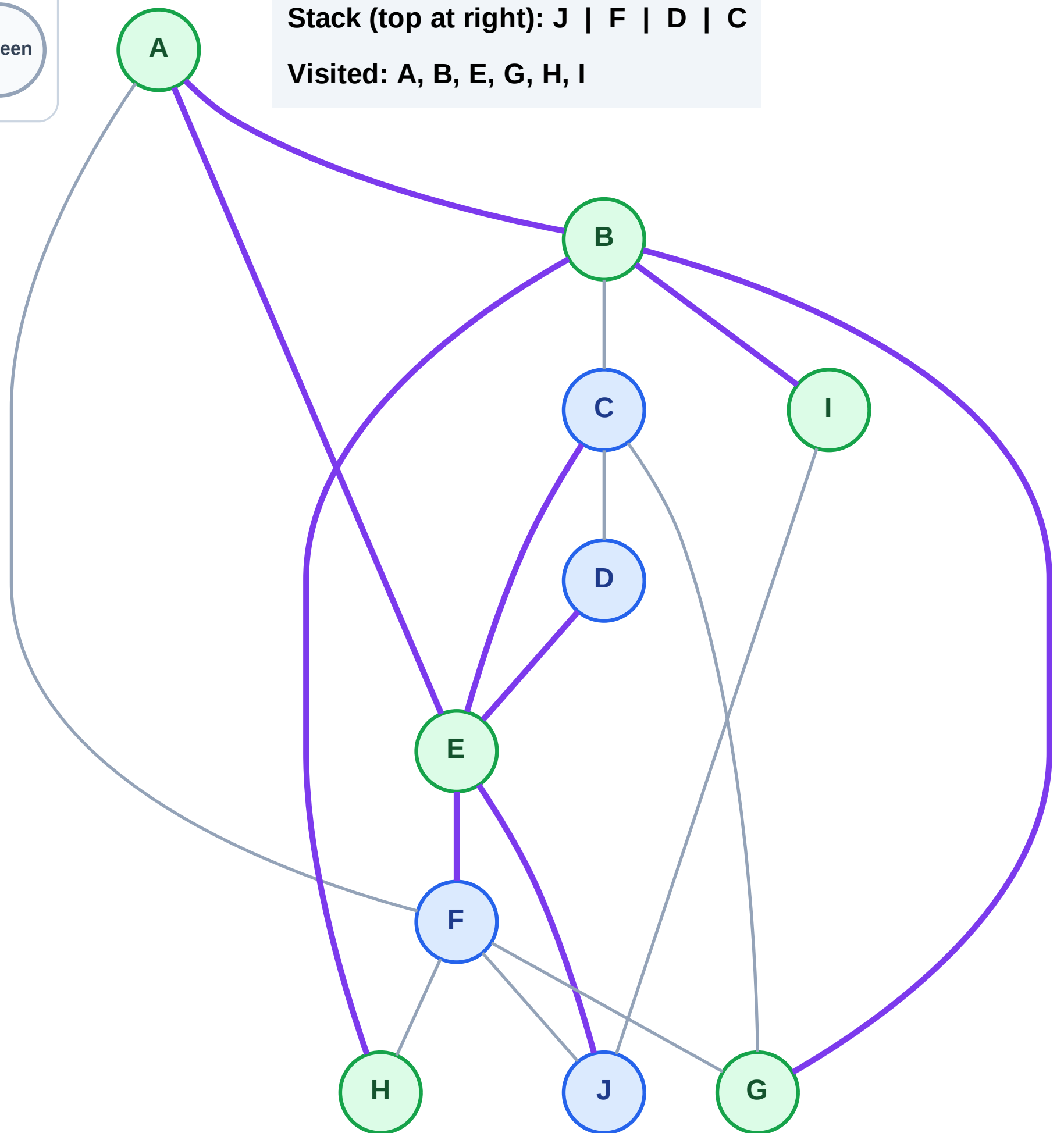
Step 13: No new neighbors from I

Legend

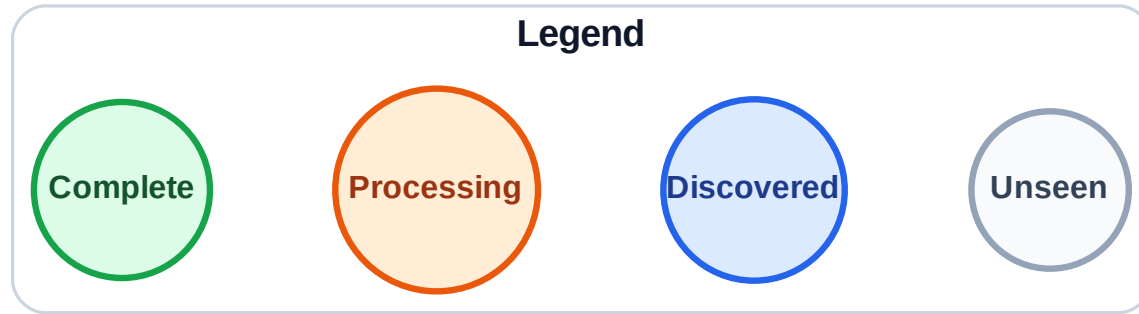


Stack (top at right): J | F | D | C

Visited: A, B, E, G, H, I

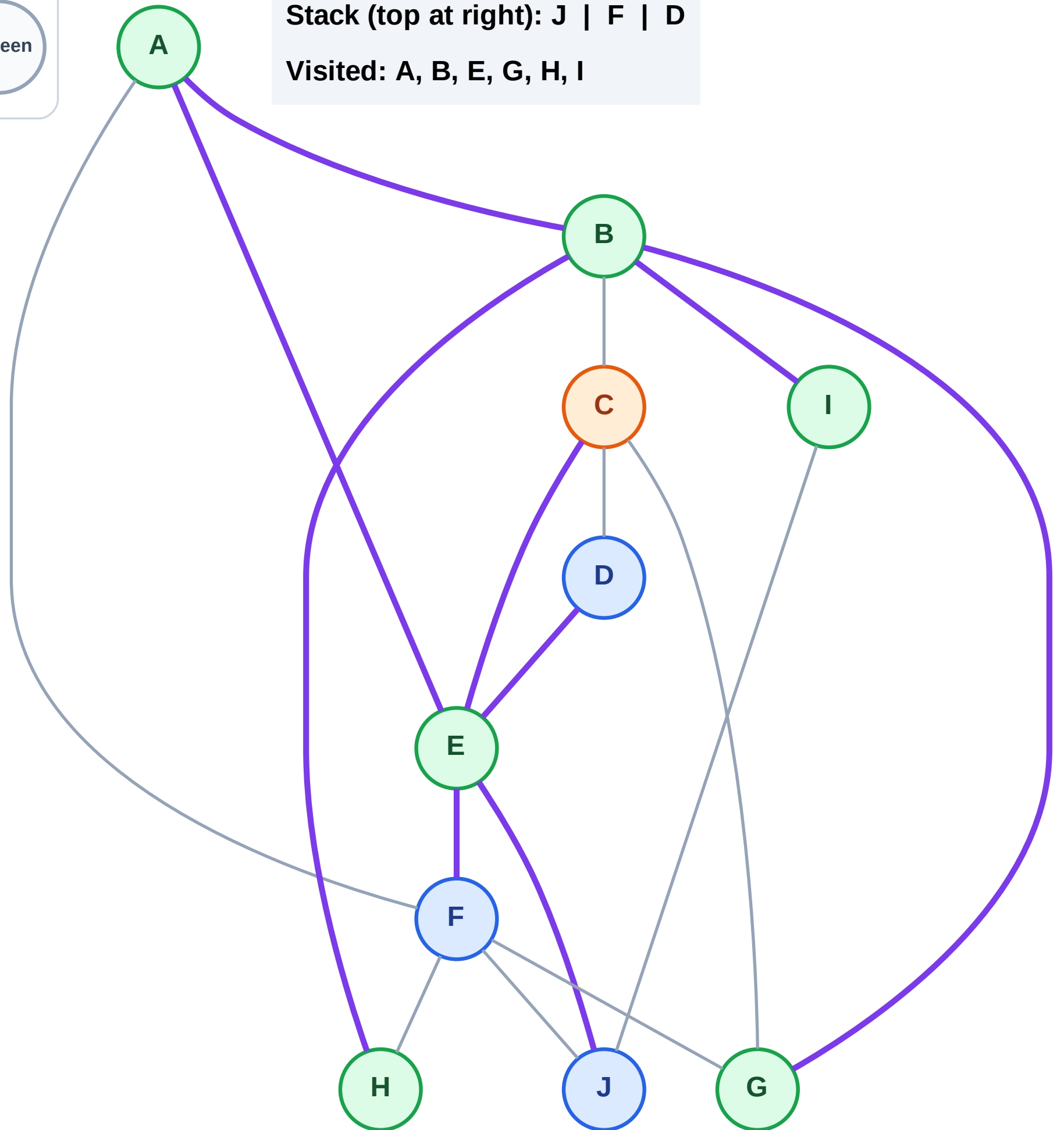


Step 14: Process node C



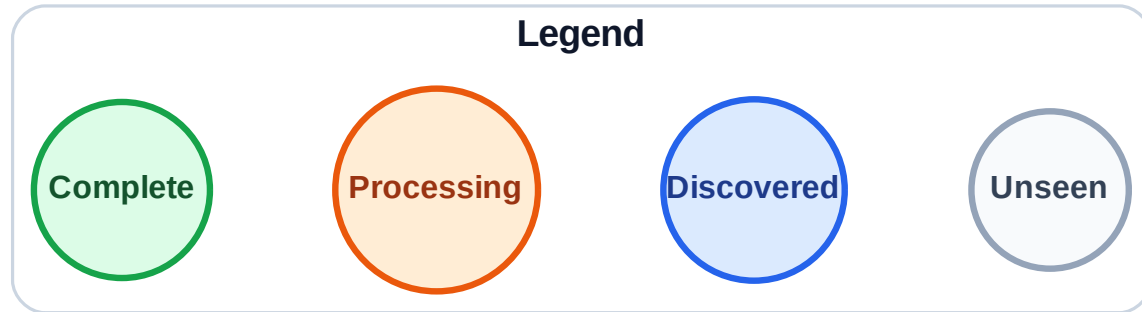
Stack (top at right): J | F | D

Visited: A, B, E, G, H, I

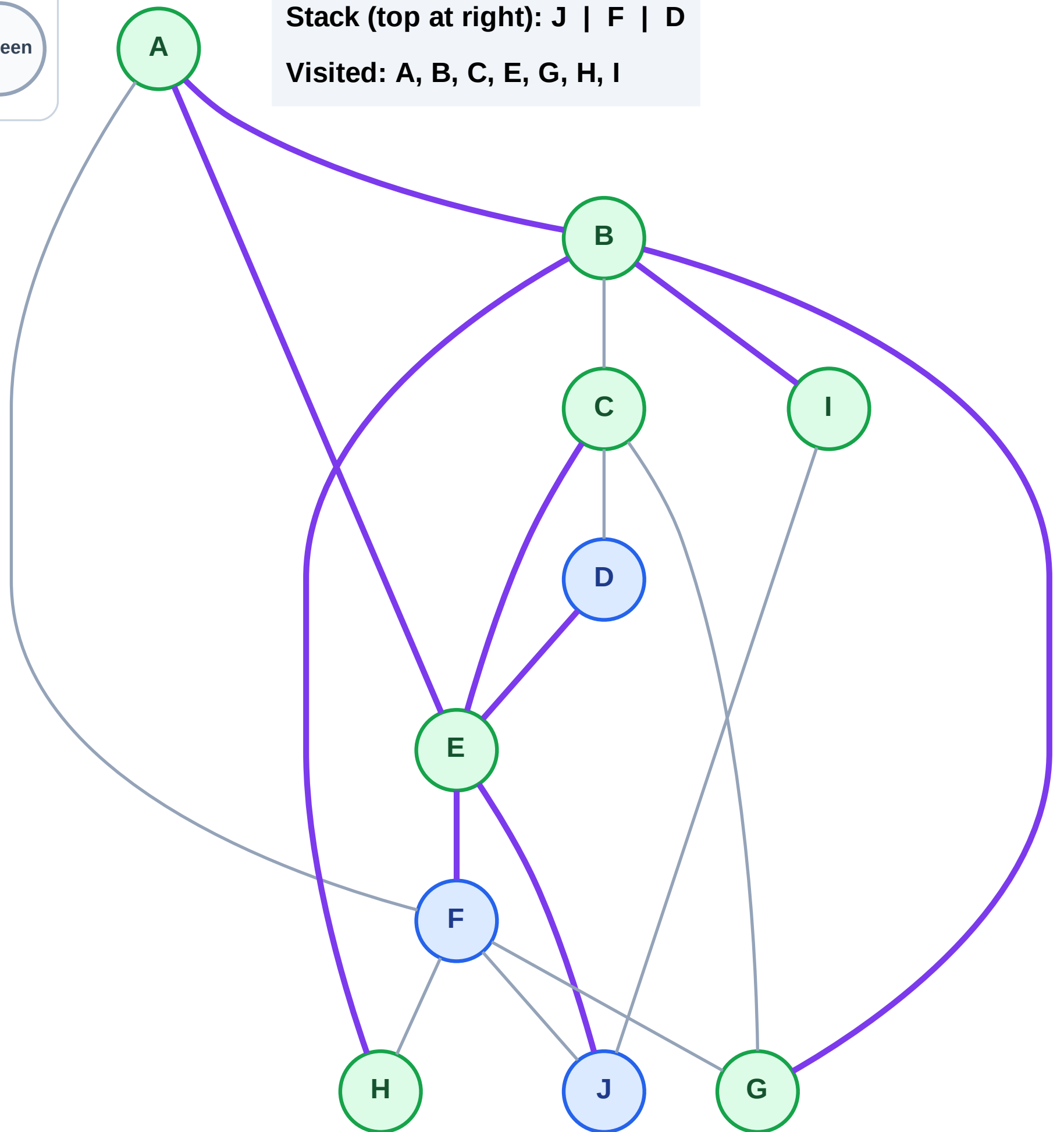


DFS Traversal

Step 15: No new neighbors from C



Stack (top at right): J | F | D
Visited: A, B, C, E, G, H, I



DFS Traversal

Step 16: Process node D

Legend



Complete



Processing



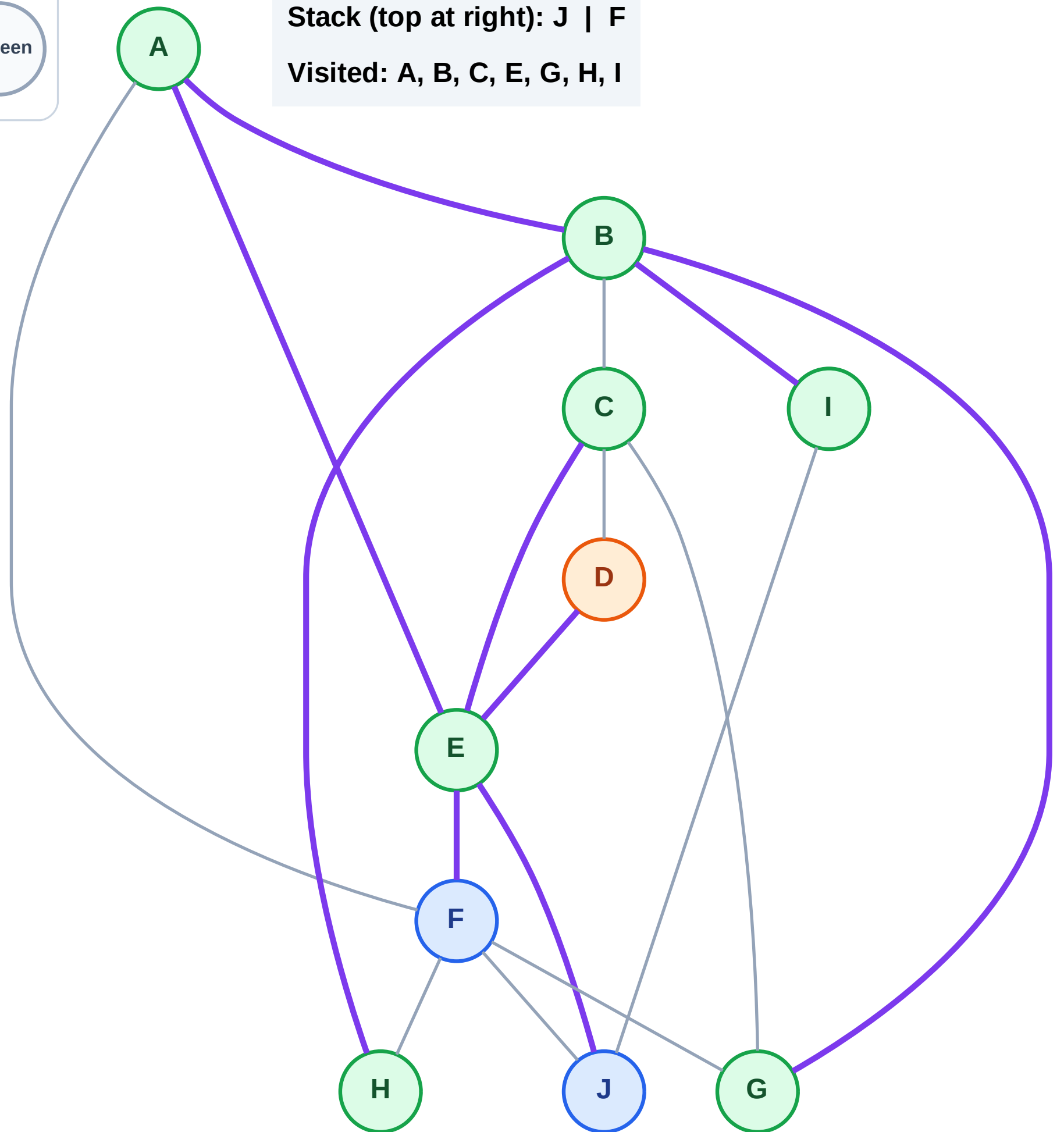
Discovered



Unseen

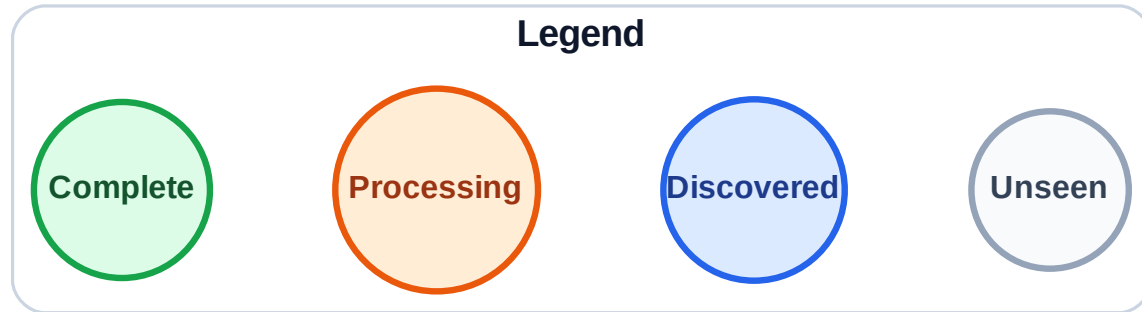
Stack (top at right): J | F

Visited: A, B, C, E, G, H, I



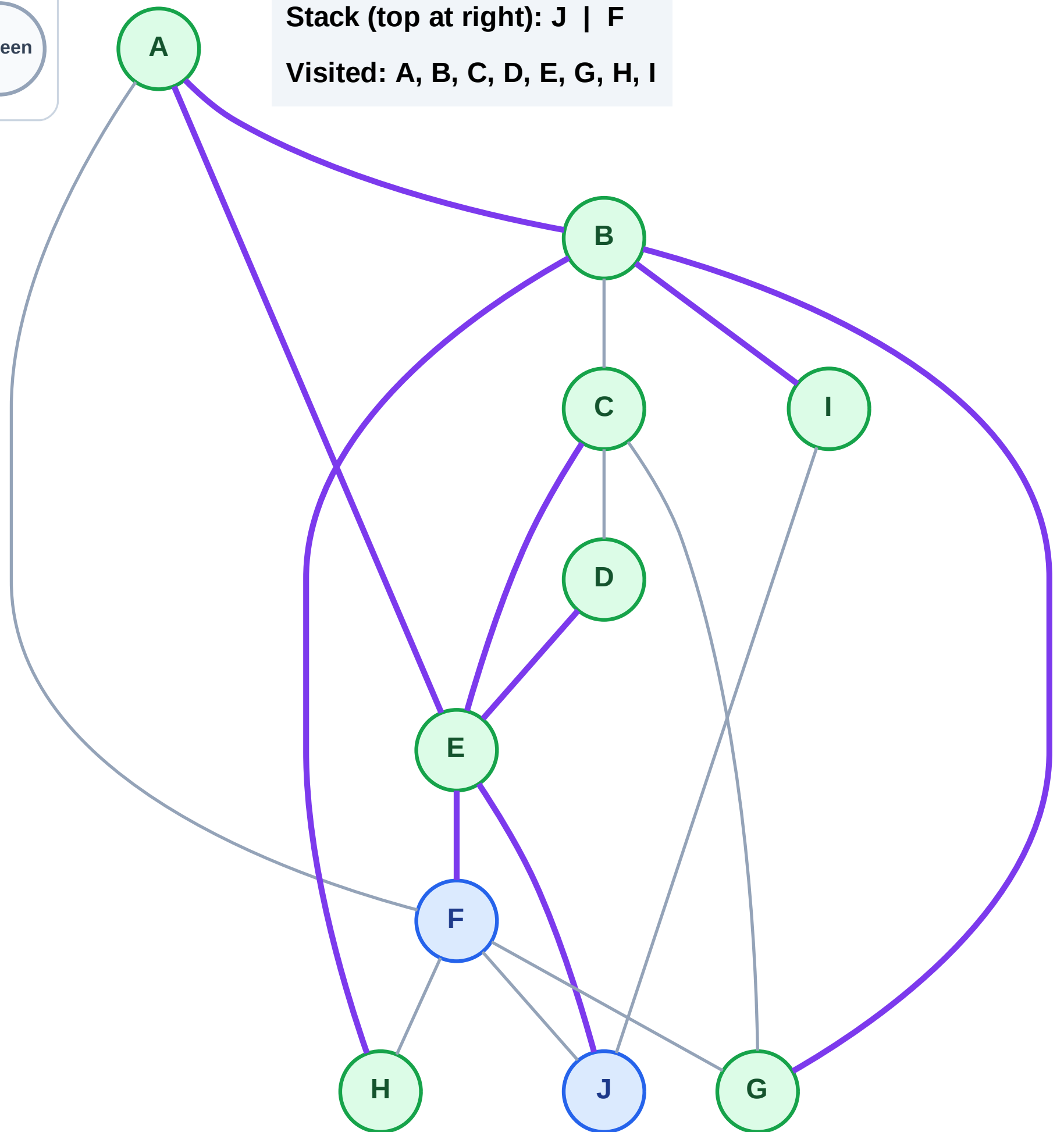
DFS Traversal

Step 17: No new neighbors from D



Stack (top at right): J | F

Visited: A, B, C, D, E, G, H, I



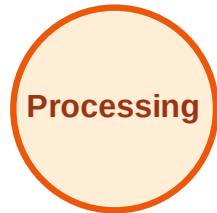
DFS Traversal

Step 18: Process node F

Legend



Complete



Processing



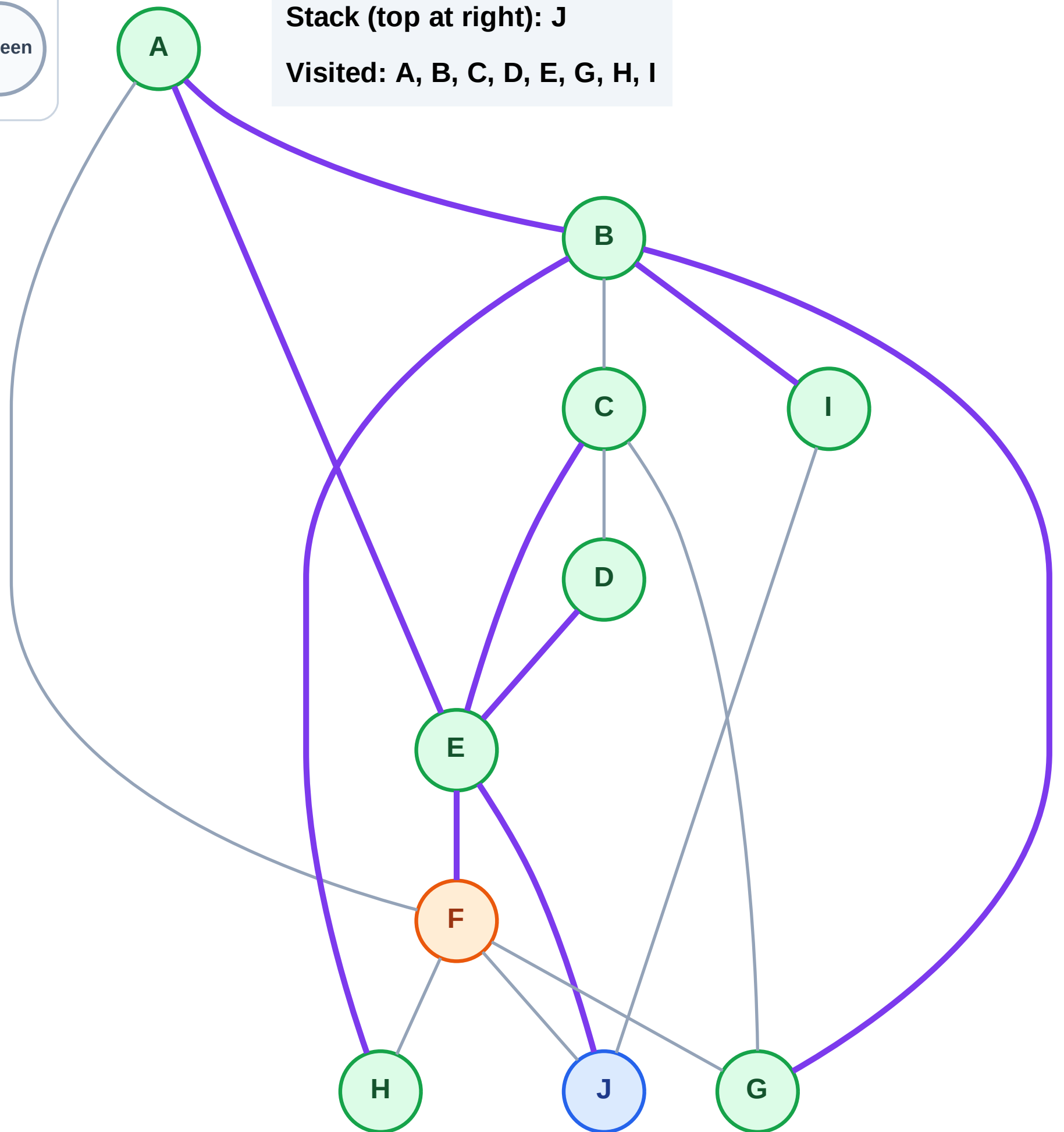
Discovered



Unseen

Stack (top at right): J

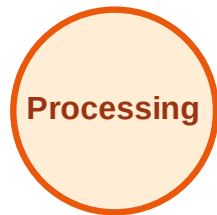
Visited: A, B, C, D, E, G, H, I



DFS Traversal

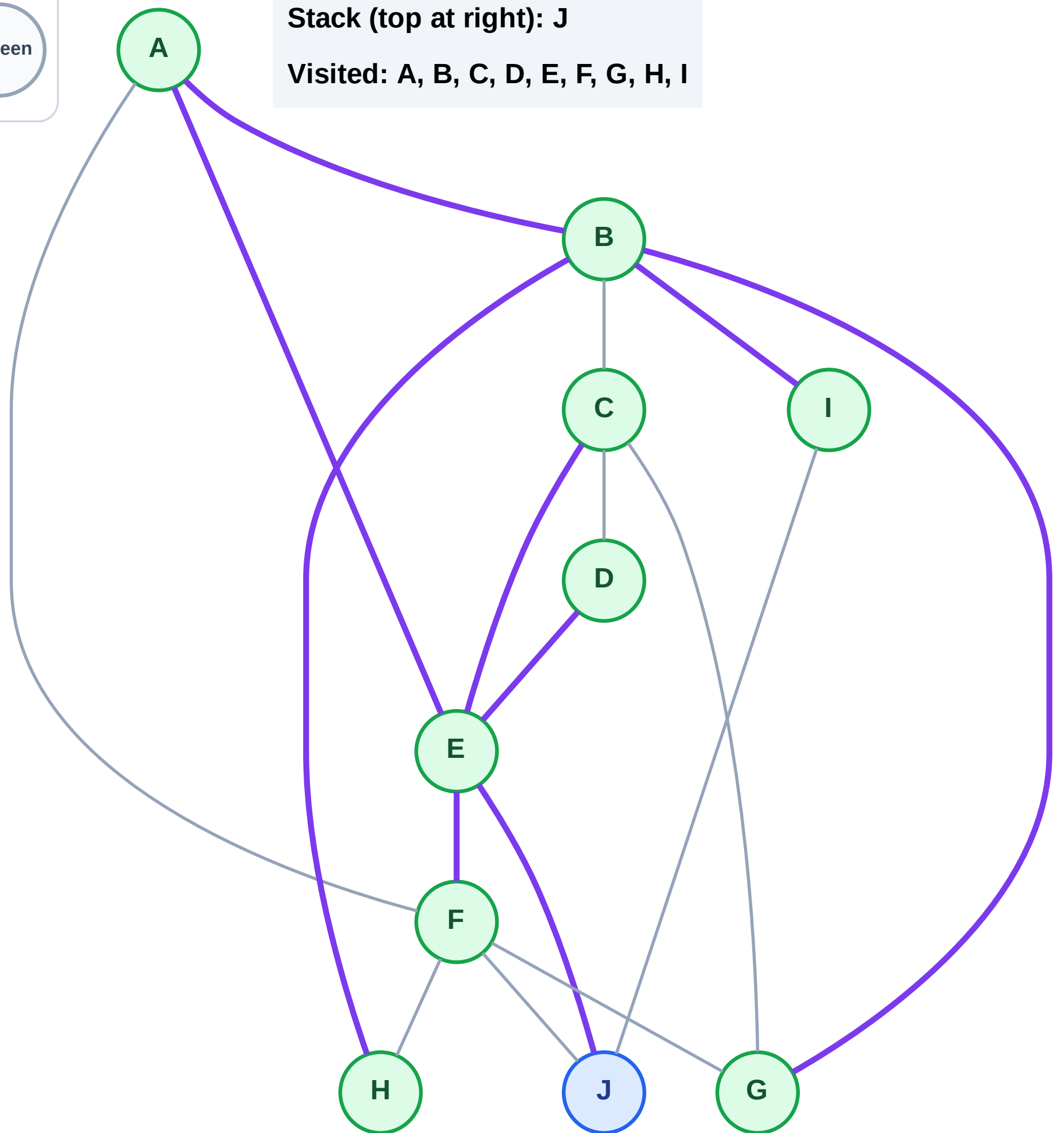
Step 19: No new neighbors from F

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 20: Process node J

Legend

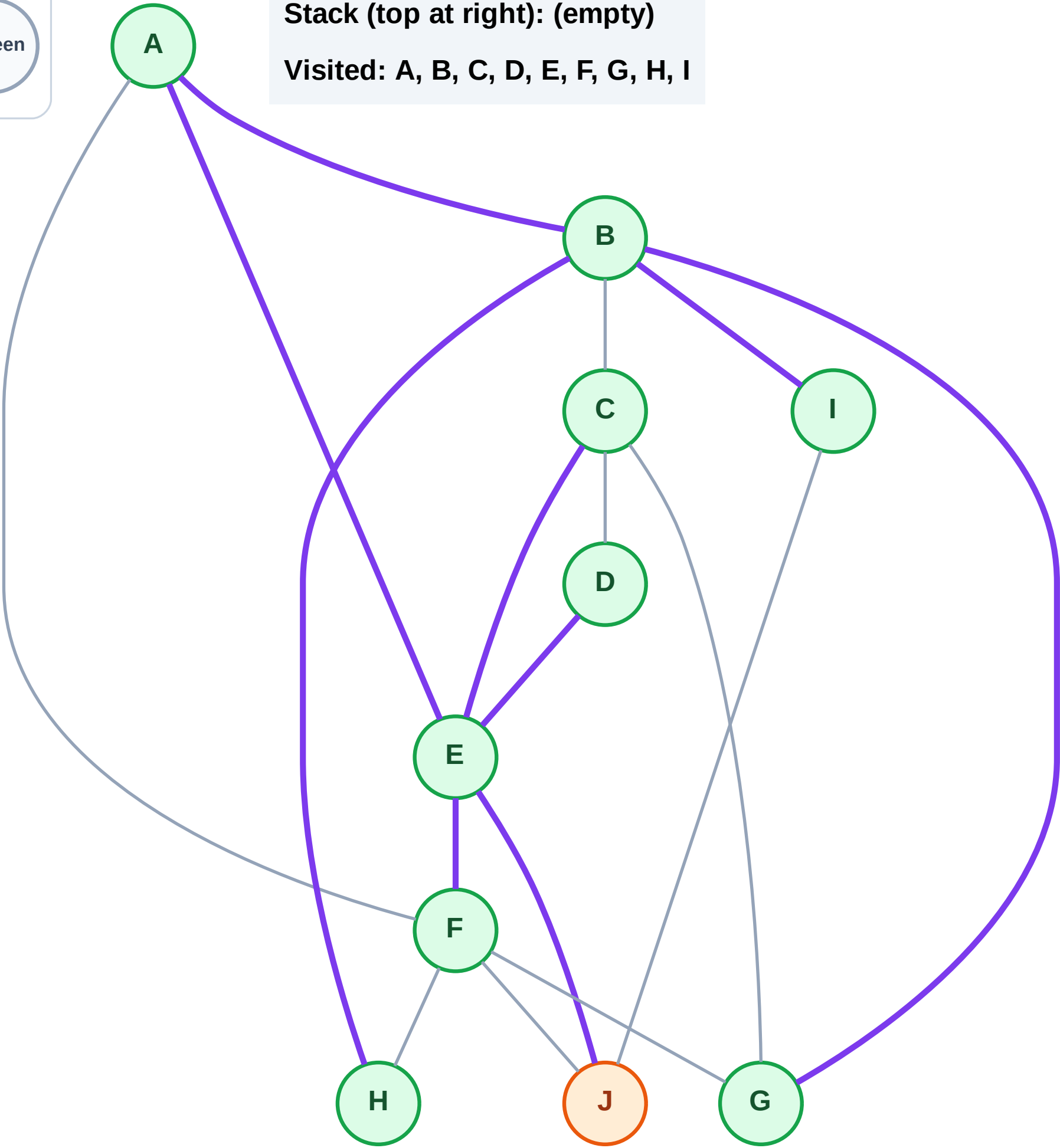
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I



Step 21: No new neighbors from J

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

